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NGA SEP, Rev B
23 September 2021



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National Geospatial-Intelligence Agency (NGA)

SYSTEMS ENGINEERING PLAN (SEP)

Revision B

23 September 2021

Abstract: (U) This document describes the NGA's conduct, management, and control of the systems engineering aspects of NGA, and provides a framework for the systems engineering of NGA's activities such as acquisition programs, projects, and studies.

Keywords: (U) Acquisition, Agile, Architecture, Configuration Management, DevOps, DevSecOps, Systems Engineering, Interface, Management, Performance, Quality, Requirements, Release, Review, Risk, SAFe, Scaled Agile Framework, Security, Service, Standards, System, Test, Transition, Verification

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1 (U) INTRODUCTION

(U) This document describes the approach to Systems Engineering followed by the National Geospatial-Intelligence Agency (NGA). The description includes principles and policies that enable NGA to strengthen systems engineering and its support to the continuing evolution of the National System for Geospatial-Intelligence.

(U) The intended audience of this Systems Engineering Plan (SEP) includes NGA systems engineers, architects (system, business, and data), program managers, and mission partners supporting systems engineering activities from the strategic to the system/segment level. The audience also includes other NSG stakeholders such as NGA management and oversight members who must understand the general engineering and technical approaches being applied in support of the NSG.

(U) This section provides a general description of the NSG and sets the context for its systems engineering practices. It also identifies key SEP related documents and information, describes the process for updating the SEP, and describes the basic structure of the document. The practices identified in this SEP are recommended at all program levels and all phases of the systems lifecycle. For the purposes of this SEP, programs are defined and inclusive of Major System Acquisitions (MSA), Integrated Program Offices (IPOs), Sensor Integration Program (SIP) Offices, Government Program Management Offices (GPMOs), Programs, Integration Projects, Segments, and Teams.

1.1 (U) The National System for Geospatial-Intelligence (NSG)

(U) The NSG is defined as “the combination of technology, policies, capabilities, doctrine, activities, people, data, and communities necessary to produce Geospatial Intelligence (GEOINT) in an integrated multi-intelligence, multi-security domain environment”. Based on this definition, the NSG includes elements that extend beyond the control of NGA. The technical approach described in this SEP applies to those elements of the NSG that NGA funds and directly manages.

1.2 (U) Technical Approach

(U) The technical approach provided in this SEP addresses activities necessary to support the life cycles of systems/segments/services/capabilities within NGA. For the purposes of this SEP, the term activity is defined to include any acquisition program, project, study, and task that involves a systems engineering function.

1.3 (U) SEP Scope

(U) This SEP is:

- a plan to guide NGA’s systems engineering practices in support of the NSG;
- intended to cover acquisition programs, regardless of type, size, length, or cost;

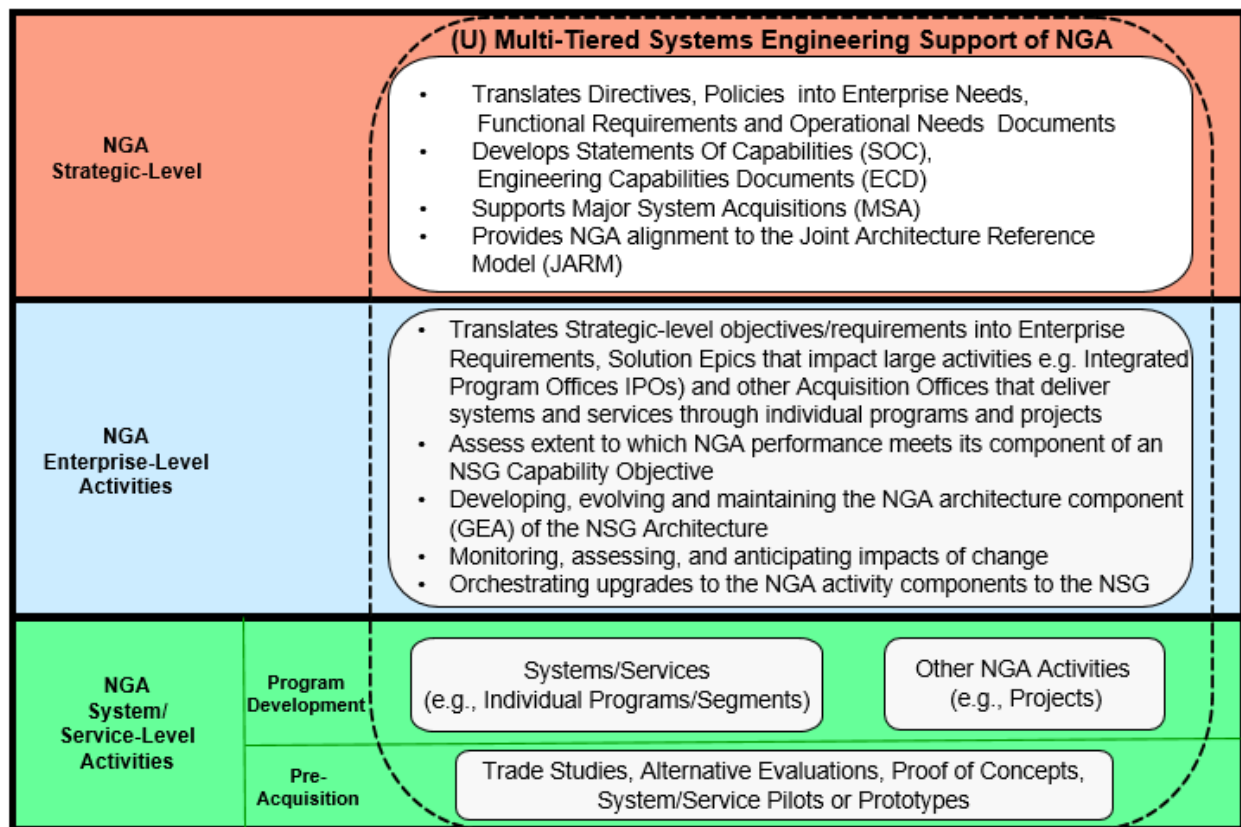
- supportive of initiatives such as service-centric orientation and agile acquisition at all levels, Enterprise to System/Segment that are pushing the NSG toward smaller elemental pieces and quicker more responsive activities.

1.4 (U) NSG Systems Engineering Activities Scope & Relationships

(U) As a System of Systems (SoS), the NSG has a life span that encompasses and transcends the lifecycle of each of the Systems/Segments/Services/Capabilities that it comprises. The Defense Acquisition Guidebook (DAG) Chapter 3 and the curriculum of the Defense Acquisition University (DAU) define processes that are applicable to both technical management systems and technical engineering. DAG/DAU technical management system processes include: technical planning, Configuration Management (CM), interface management, technical data management, requirements management, risk management, technical assessment, and decision analysis. The DAG/DAU technical engineering processes are: stakeholder requirements definition, requirements analysis, architecture design, development, integration, verification, validation, and transition which may be applicable to both traditional and agile engineering and acquisitions initiatives. These processes should be tailored to meet the program's needs and development methodology.

(U) Figure 1.4-1 illustrates a multi-tiered approach to systems engineering; specifically, how enterprise-level systems engineering activities relate to the execution of the program-level systems engineering activities. All activities use systems engineering and technical management processes, but the depth and focus of the application of those processes is a key differentiator of large versus small activities. It is important to understand the size, scope, and significance of the program in order to understand how to tailor systems engineering processes to achieve maximum flexibility and agility in achieving program goals/objectives and delivering capabilities to the user in the minimum amount of time necessary.

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Figure 1.4-1 (U) A Multi-Tiered Approach to Systems Engineering

1.5 (U) NGA Acquisition Lifecycle

(U) Figure 1.5-1 highlights the five sub-phases of the NGA Standard Acquisition Process (i.e., from technology development to operations and sustainment) and aligns technical processes to an agile development cycle or a traditional lifecycle.

(U) Technology Development. Both agile and traditional development activities begin with a technology development phase that includes such activities as an initial concept definition (e.g. Concept of Operations (CONOPS), Statement of Capabilities (SOC)) preliminary requirements are defined and Analysis of Alternatives (AoAs) and prototyping activities are conducted.

- (U) Engineering: In agile development, NGA uses Solution Epics (SEs) to capture development efforts at an Enterprise-level. An SE consists of multiple SE Minimum Viable Product functions (MVPf(s)). The SE MVPf(s) are further decomposed into Program Epics (PE). There is no set timeframe in developing these artifacts (section 3.1 offers further discussion on SEs, PEs, and MVPf(s)). Traditional development requirements can be captured in either a System

Requirements Document (SRD), Capability Oriented Requirements (COR), or Design Document (DD).

- (U) Manufacturing, Development, Integration and Test: Both agile and traditional development conduct these activities, the major difference is when capabilities are going to be available to the user. Agile manufacturing, development, integration, and testing are iterative processes where capabilities are released every 8 to 12 weeks. In traditional manufacturing and development these activities occur in a serial manner and once completed, capabilities are delivered to the Integration and Test team for final testing. The time duration for these activities can be up to a year (or greater) in duration.
- (U) Deployment: In both agile and traditional development, after the developer has completed internal Manufacturing, Development, Integration and Testing of new capabilities, both are delivered to the NGA Test Organization (NTO) for independent Testing and Evaluation (T&E) of changes to the NSG Enterprise baseline prior to operational deployment. It is at the NTO where Authority to Operate (ATO) is granted.
- (U) Operations and Support: Regardless of the systems engineering method used, once capabilities have completed NTO testing and achieved ATO, capabilities are deployed into operations by User Facing Services (UFS) and once accepted for use by the Missions Owners, the capability is now in Sustainment.

(U) The purpose of the NGA Enterprise Readiness (ER) Review process is to facilitate efficient and effective Acquisition Readiness reviews that assess whether segments, systems, services, capabilities, and/or applications are ready to be moved through the various milestones in the NGA Acquisition Lifecycle, including into or out of the operational environment. The ER process is tailorable to include the type and number of program reviews that may be required and documentation required at each review.

(U) For MSAs, NGA must adhere to the milestones as prescribed in IC Directive 801, and deliver the required documentation for each milestone (Appendix C). The three high-level milestones are:

Milestone A: Occurs prior to the beginning of the Technology Development phase of the acquisition lifecycle.

Milestone B: Occurs upon completion of the Technology Development phase and prior to entering the Engineering and Manufacturing phase of the acquisition lifecycle.

Milestone C: Occurs at the end of the Engineering and Manufacturing phase and prior to the beginning of the Production and Deployment phase.

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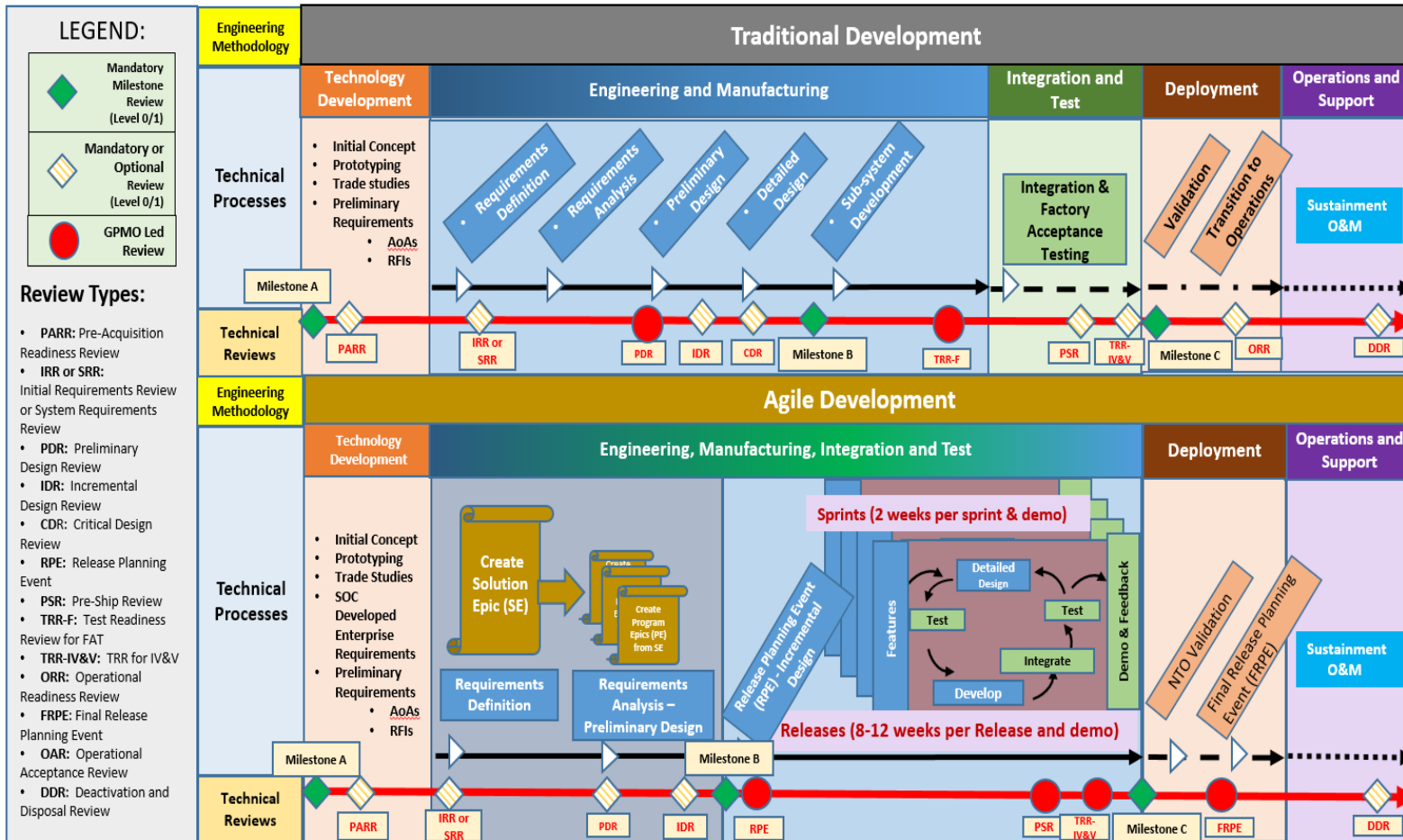
(U) The execution of these milestones is governed by the criteria and thresholds established by ICD 801 and DoD Directive 5000.01.

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Figure 1.5-1 (U) Examples: NGA Acquisition Lifecycle Activities

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(U) Enterprise-level Systems Engineers are integral to this process, as they must evaluate, monitor, and orchestrate the acquisition and upgrade of multiple systems/segments/services at the different stages of the systems engineering lifecycle which covers initial concept development through deployment, sustainment, and retirement.

1.6 (U) De-confliction of Priorities

(U) Programs and their suppliers need to be attentive and timely to users' needs throughout the community, priorities may become conflicted on occasion require de-confliction. De-confliction of priorities must take into account mission, technical, cost, and scheduling impacts to the program.

(U) In the event that a program and its internal suppliers are unable to resolve prioritization to an acceptable level, the prioritization issue will be escalated to CIO-T senior leadership and senior level user representatives for joint adjudication. For engineering efforts, in instances where programs and their external suppliers are unable to resolve priority disagreements at the program or IPO levels, issues can be escalated to the Engineering Review Board (ERB), as applicable.

1.7 (U) SEP Configuration Control

(U) The NGA Configuration Control Board (CCB) is the approval authority over future versions/updates to the SEP. However, technical ownership of this document falls within the responsibility of the NGA Chief Engineer.

2 (U) APPLICABLE DOCUMENTS

(U) The information provided in this SEP reflects enterprise-level systems engineering guidance from a variety of information sources. These sources include Department, Community, and Agency-level guidance documents. Overall, NGA systems engineering activities are influenced by direction and guidance from a variety of internal/external policies, processes, procedures, standards, and frameworks. The sources for this information include government, industry, academic, and professional sources.

GEA	GEOINT Enterprise Architecture version 4.5	Jun 2019
NCMP	NGA Configuration Management Plan	Jun 2021
NERP	NGA Enterprise Requirements Process	May 2021
NGA ADE	NGA Acquisition Design Elements	Aug 2020
NGA ERP	NGA Enterprise Readiness (ER) Process Document	Jan 2020
NGA TEMP	National System for Geospatial-Intelligence (NSG) Enterprise Test and Evaluation Master Plan	Jun 2018
NSMP	NGA Schedule Management Plan	Sep 2020
Tier 2 CM Procedure	Tier 2 Configuration Management (CM) Procedure	May 2018
DAG	Defense Acquisition Guidebook	Sep 2020
DoDD 5000.01	Department of Defense Directive 5000.01, The Defense Acquisition System, May 2003 incorporating Ch. 2	Aug 2018
DoDI 5000.2	Department of Defense Instruction 5000.2, Operation of the Defense Acquisition Framework	Jan 2020
DoDI 5000.87	Department of Defense Instruction 5000.87, Operation of the Software Acquisition Pathway	Oct 2020
DoD DE Strategy Guide	Department of Defense Digital Engineering Strategy Guide	Jun 2018
CNSSI 1253	Committee on National Security Systems (CNSS) Instruction No. 1253	Mar 2014

ICCR (ICD115)	Intelligence Community Capabilities Requirements (ICCR) Process (ICD 115)	Dec 2012
ICD 503	Intelligence Community Information Technology Systems Security Risk Management	Sep 2008
ICD 801	Intelligence Community Directives Number 801	Aug 2009
IC PAG	Intelligence Community Program Architecture Guide version 2	Feb 2013
JARM	Joint Architecture Reference Model version 5.0	Mar 2016
ODNI ICS 801-01	Intelligence Community Standard 801-01, Major Systems Acquisition	Jun 2013
ITRMP	Information Technology Risk Management Process	Apr 2021

3 (U) ENGINEERING ELEMENTS AND PROCESSES

(U) This section provides an overview of various components of systems engineering and related processes. Detailed descriptions and instructions for the implementation of each process within NGA is provided via attached enclosures and/or appendices, as appropriate.

3.1 (U) Enterprise Release Management (ERM)

(U) ERM orchestrates and integrates engineering activities and resources across multiple interdependent system delivery life cycles in order to meet project objectives across the NSG Enterprise.

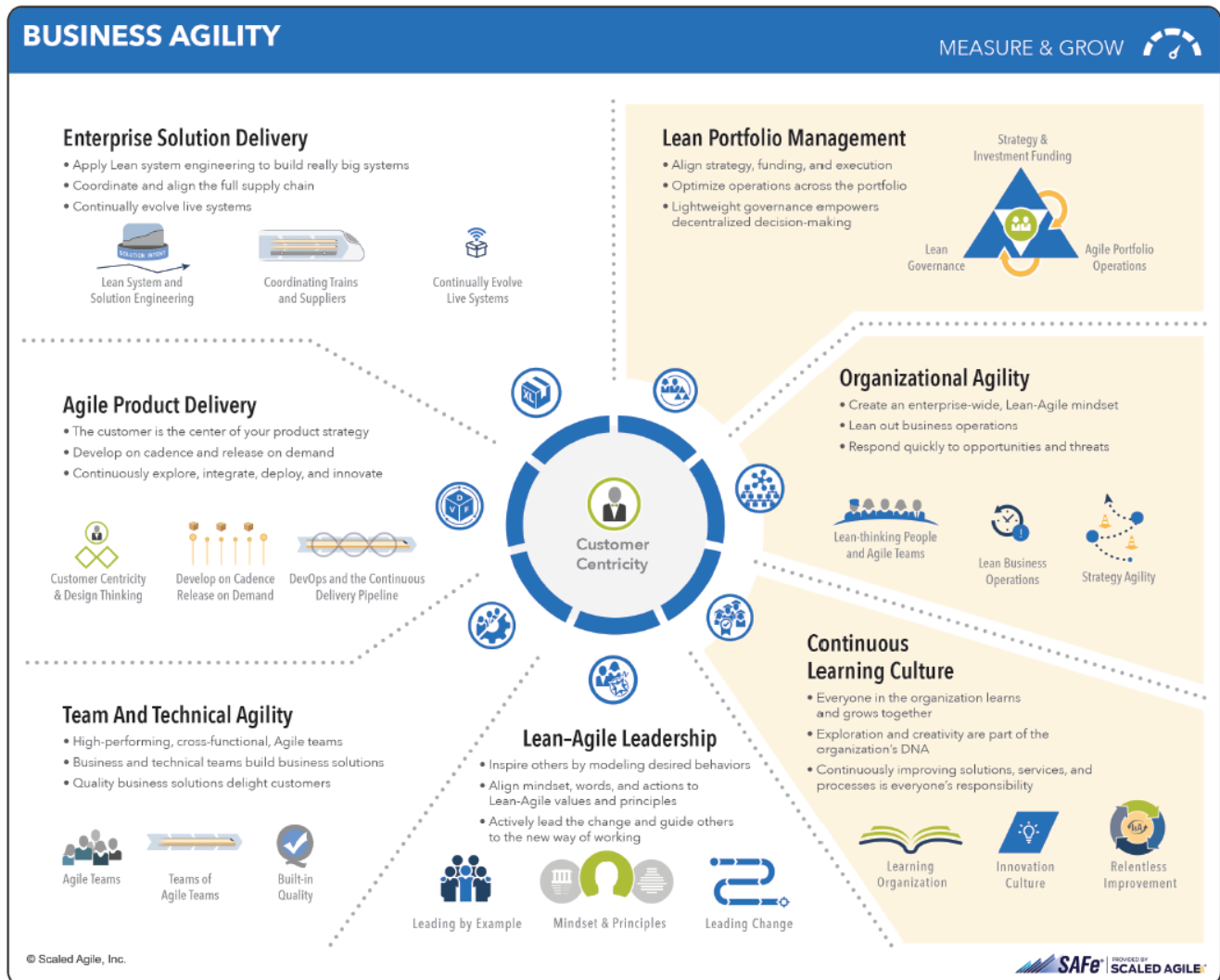
(U) In order to achieve this, ERM facilitates and supports system development and integration efforts across all programs. ERM uses the current version of the Scaled Agile Framework® (SAFe®) for Lean Enterprises (as of this writing SAFe v. 5.1) as the basis to enable the NSG to conduct enterprise-level Agile development activities that deliver defined objectives. ERM closely integrates with Requirements, Architecture, Digital Engineering, Schedules, and other engineering activities to actively anticipate, de-conflict, monitor and manage cross-contract dependencies and schedules for enterprise level programs. These activities ensure mission alignment and synchronization across projects, programs, and segments to rapidly deliver capabilities that fulfill mission needs.

(U) The goals of ERM are to provide:

- Enterprise Agility
- An Enterprise-level visibility of project/program development activities
- A common look and feel of project/program artifacts
- Top-to-bottom traceability of development activities – from the Enterprise to contract development level
- Normalized Project and Program roadmaps

3.1.1 (U) Enterprise Agility

(U) NGA is moving away from traditional development and acquisitions processes in favor of agile. As such, ERM follows and recommends the implementation of SAFe which “integrates the power of Lean, Agile, and DevOps into a comprehensive operating system that helps enterprises thrive in the digital age by delivering innovative products and services faster, more predictability, and with higher quality.” Figure 3.1.1-1 provides an introduction to the seven core competencies of the Lean-Agile Enterprise.



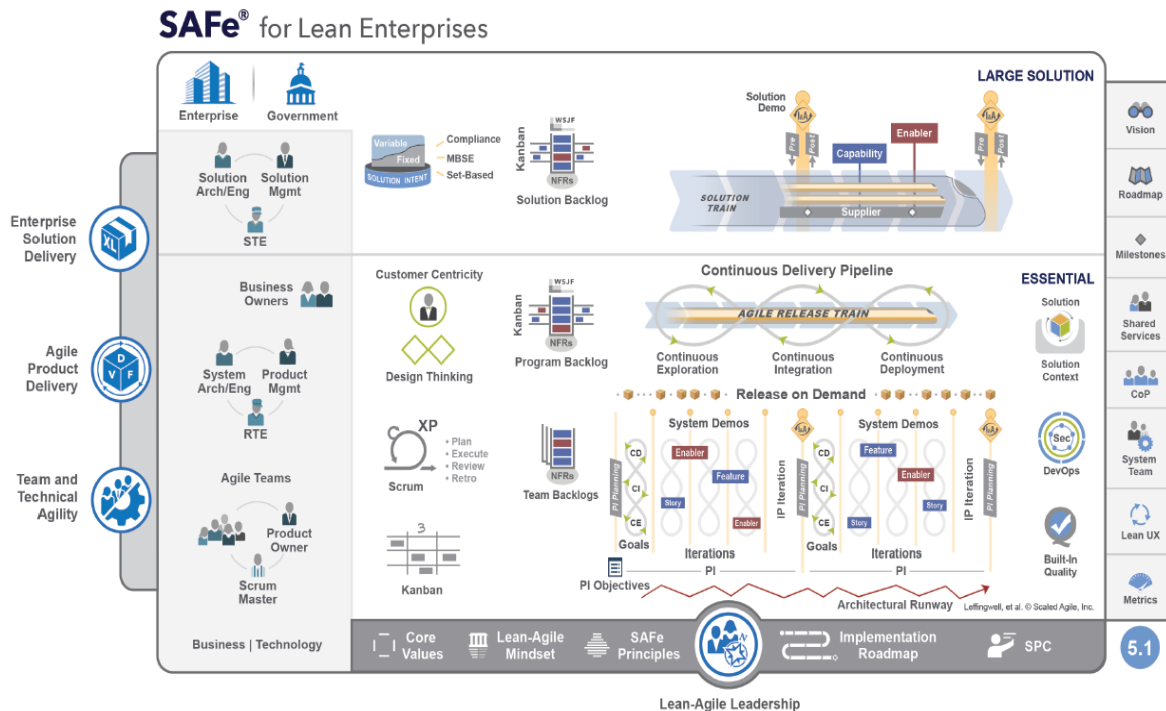
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Figure 3.1.1-1 (U) Scaled Agile Framework® Overview

(U) ERM Enterprise Agile activities include:

- Promotes the harmonization and implementation of Enterprise-Level agile practices at scale such that different programs can interact with one another in a standardized and uniform manner
- Provides mentorship to teams that are new to working within Enterprise-Level agile constructs
- Provides guidance to programs as they develop their SEs and PEs to align with Enterprise Agile best practices while remaining in compliance with both the ERM Epic Framework and the Mission Needs Taxonomy.

(U) To support the needs of enterprise-level programs and thus create harmonization across these programs, ERM has adopted the “Large Solution” SAFe configuration (Figure 3.1.1-2) which provides the requisite positions, components, processes, and workflows necessary to allow programs to meet their objectives and mission needs. The ERM team works with program partners to enhance their introduction to Large Solution SAFe; provides introductory training (Introduction to ERM, artifact generation, tool set-up and usage, etc.); and mentorship as needed as the program progresses through its lifecycle. The Large Solution SAFe Configuration also supports the top level Solution Epic. Epics can best be defined as a container for a solution development initiative large enough to require analysis, the defining of a Minimum Viable Product (MVP), and financial approval before implementation. The Large Solution configuration also establishes the organizational construct of a “Solution Train” that is used to build large and complex Solutions (often as an SoS) that require the coordination of multiple Agile Release Trains (ARTs), as well as the contributions of both internal and external suppliers (e.g., programs, Acquisition Offices (AOs), Mission partners, etc.). For additional information about Solution Trains and their subordinate agile teams please refer to: <https://www.scaledagileframework.com/solution-train/>.



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Figure 3.1.1-2 (U) Large Solution SAFe® Configuration**3.1.2 (U) Enterprise-Level Visibility**

(U) Enterprise-level visibility is achieved by working closely with partners, conducting periodic briefings, chairing working group meetings, and providing the ERM space on NGA's Enterprise instance of the Atlassian Confluence™ tool. ERM's Confluence site provides visibility into ERM's mission and the development activities of programs by providing a program specific space for each program partner to create and store program artifacts that support the generation of SEs and PEs. Additionally, the ERM Confluence site serves as a curated knowledge repository that includes access to documentation, templates, automation tools, useful "how-to's," and team contact information.

3.1.3 (U) Normalized Artifacts

(U) NGA programs are responsible for managing their requirements baseline at both the enterprise level and program levels. To do so, agile programs use SEs and PEs for this purpose, as depicted in the epic framework Figure 3.1.3-1. A repository of templates and best practices for creating normalized can be found in the ERM Confluence site.

(U) Solution Epics: An SE captures the customer facing capabilities to be delivered and is managed by the Epic owner. SEs are typically cross-cutting and capture development efforts needed to realize mission focused initiatives. The SE is an enterprise-level artifact used in agile development, and it captures mission needs in a development initiative large enough to require analysis; consists of multiple SE Minimum Viable Product functions (MVPf(s)), and includes success criteria. SE MVPf(s) are the minimal viable functions the user has requested that are required to deliver a minimal, viable solution. The format and content of SEs is the Lean Business Case (LBC), which is tailored from the SAFe LBC. The planning horizon for an SE is up to five years. It is important to note that SEs are CCB controlled.

(U) Program Epics: A PE is designed for a single development activity to accomplish SE MVPf(s). Like SEs, PEs also have MVPf(s), and multiple PE MVPf(s) may be required to deliver a SE MVPf. These PE MVPf(s) are specific tasks the program team must execute to satisfy their allocated portion of the SE MVPf(s). When requirements are not traced via an agile construct, PE MVPf(s) are traced back to the appropriate program requirements found in a DD, COR, or SRD. The planning horizon for a PE is up to two years. PEs are program controlled.

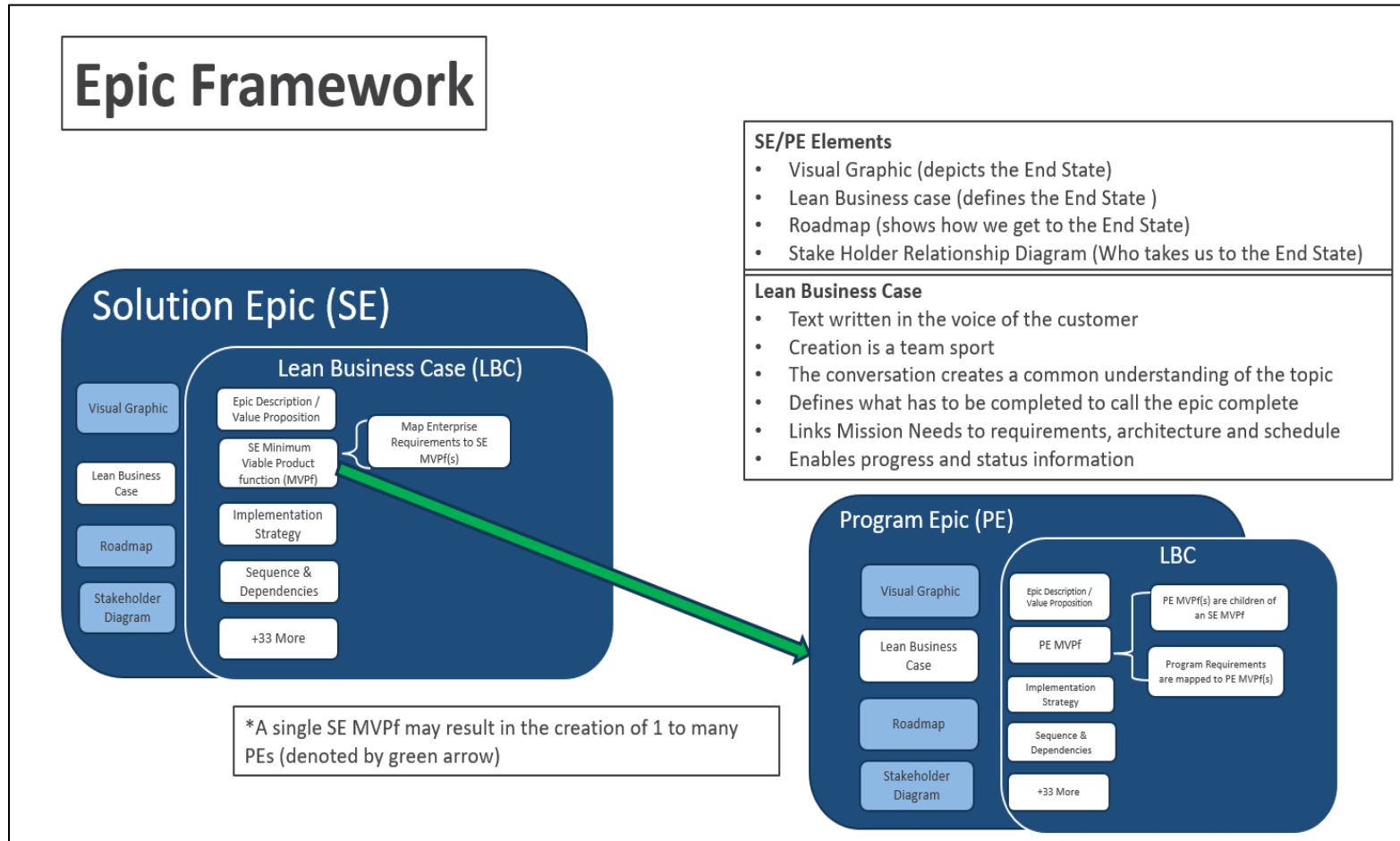


Figure 3.1.3-1 (U) Epic Framework

(U) Epic creation is a collaborative effort, meaning the Product Owner/Product Manager (POPM), Architects, Engineers and other stakeholders are in the same room at the same time developing the content.

(U) The Framework for both SEs and PEs is the same, and both are comprised of the following four artifacts:

1. End State Graphic
2. Lean Business Case
3. Roadmap
4. The Stakeholder Diagram

3.1.3.1 (U) End State Graphic

(U) The End State Graphic describes the end-state vision of the SE/PE. This is usually an Operational View (OV) -1, or an OV-2. The diagram is descriptive enough to convey the desired outcome of development efforts. It is recommended this artifact be created first.

3.1.3.2 (U) Lean Business Case

(U) The LBC defines and describes what needs to be done to reach the end-state. The LBC text is written in the voice of the customer. This creates common understanding and ownership amongst the stakeholders.

(U) The majority of the fields in the LBC template are self-explanatory; however, four sections warrant additional discussion.

- (U) Epic Description/Value Proposition: Once the End State graphic is created, the Epic Description/Value Proposition is developed, using the following format:
 - [For] users of the proposed capability being developed;
 - [Who] who performs what mission;
 - [Our] the proposed solution name;
 - [Is A] how we are going about providing it;
 - [That] describes how the proposed solution provides value and is;
 - [Unlike] unlike the current solution/capability; and
 - [Our Solution] describes how the proposed solution is better.
- (U) Minimum Viable Product functions: The MVPf(s) define what is necessary for the SE/PE to be complete. The statements are again in the voice of the customer (e.g., mission owner); describing what the user needs. The MVPf(s) are not solution, engineering, or architecture statements. For SE MVPf(s) only, the MVPf(s) are to be mapped to appropriate NGA Enterprise Requirements, as they

are created based upon the NGA Enterprise Requirements developed from strategic documentation. Program Epic MVPf(s) are mapped from the program COR, RD, or DD requirements, if applicable.

- (U) Incremental/Implementation Strategy: This defines how the epic will be implemented, e.g., capabilities will be released every two weeks (sprints), and shippable increments (Releases) will occur every 8 to 12 weeks.
- (U) Sequencing and Dependencies and Key Partners: Sequencing ensures that tasks are scheduled such that tasks are not begun until one or more other conditions, events, or tasks have occurred, have begun, or have been completed. These conditions, events, or tasks are Dependencies. Sequencing of events/tasks and dependencies are to be depicted on the Epic Roadmap.

3.1.3.3 (U) Roadmap

(U) The Roadmap depicts how to arrive at the end state over time. It is a high-level schedule of major milestones and events that show when major activities occur and when capabilities are developed that lead to the end-state. The roadmap also depicts dependencies that may impede development efforts. While the roadmap is dynamic and is refined as the effort progresses, If the SE has an RFC and the Roadmap changes, a change is needed to the RFC.

(U) Much of the information that is used to graphically depict the roadmap reflects information found in several fields within the LBC. Specifically, the following LBC fields will influence the roadmap graphic:

- Estimated Development Timeline
- Start Date
- Completion Date
- Incremental Implementation Strategy
- Sequencing and Dependencies + Key Partners
- Milestones or Checkpoints

3.1.3.4 (U) Stakeholder Relationship Diagram

(U) A Stakeholder Relationship diagram is a visual description of who will take the program to its end state. This diagram illustrates the “major players” that must work together to deliver this end state and how they are related to each other. Stakeholders, by definition, can be internal or external and are those people or programs that have an interest in or will be impacted by the capabilities provided within the solution. This diagram could be a Systems View (SV-1) or an OV-4.

(U) Much of the information that is used to graphically depict the Stakeholder Relationship Diagram will have been identified in several fields within the Roadmap. Specifically, the following LBC fields will influence this graphic:

- Sponsors
- Users and Markets Affected
- Impact on Products, Programs and Services
- Sequencing and Dependencies + Key Partners

3.1.4 (U) Epic Creation

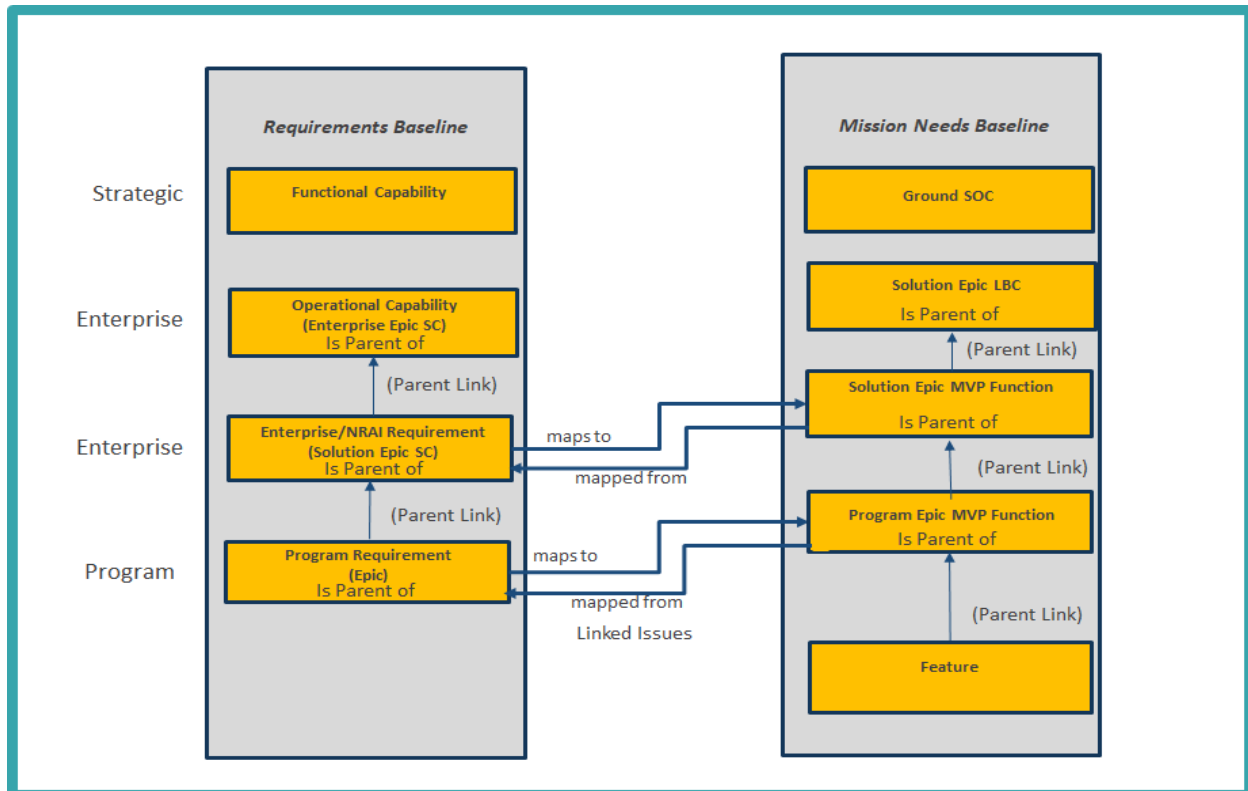
(U) The tool used in NGA to create SEs and PEs is Confluence. Confluence provides a collaborative environment that tracks changes (and who made these changes) to the SE or PE. The ERM Team provides space on its Confluence page for programs to develop their SEs and PEs. The approved SE/PE template is pre-loaded (by the ERM Team) into the project/program section to ensure consistent artifacts across the enterprise. The ERM Confluence site also provides information on how to create SEs and PEs. If the SE and/or PE is to be baselined via the Request for Change (RFC) process, ERM has a tool to export the complete SE or PE (End State Graphic, LBC, Roadmap, and Stakeholder diagram) into a Microsoft Word® or Adobe® PDF document type for inclusion in the RFC package.

3.1.5 (U) Enterprise to Contract Traceability

3.1.5.1 (U) Requirements and Mission Needs Taxonomy

(U) NGA's Enterprise Jira database is used to achieve Enterprise-level to contract-level development traceability. The basic Requirements and Mission Needs taxonomies (Figure 3.1.5.1-1) were developed to show how requirements and mission needs are related at the Enterprise and Contract level. The Requirements and Mission Needs Taxonomy was developed to provide the structure for entering and linking SEs, SE LBCs, PEs, PE MVPf(s), and feature data into Enterprise Jira.

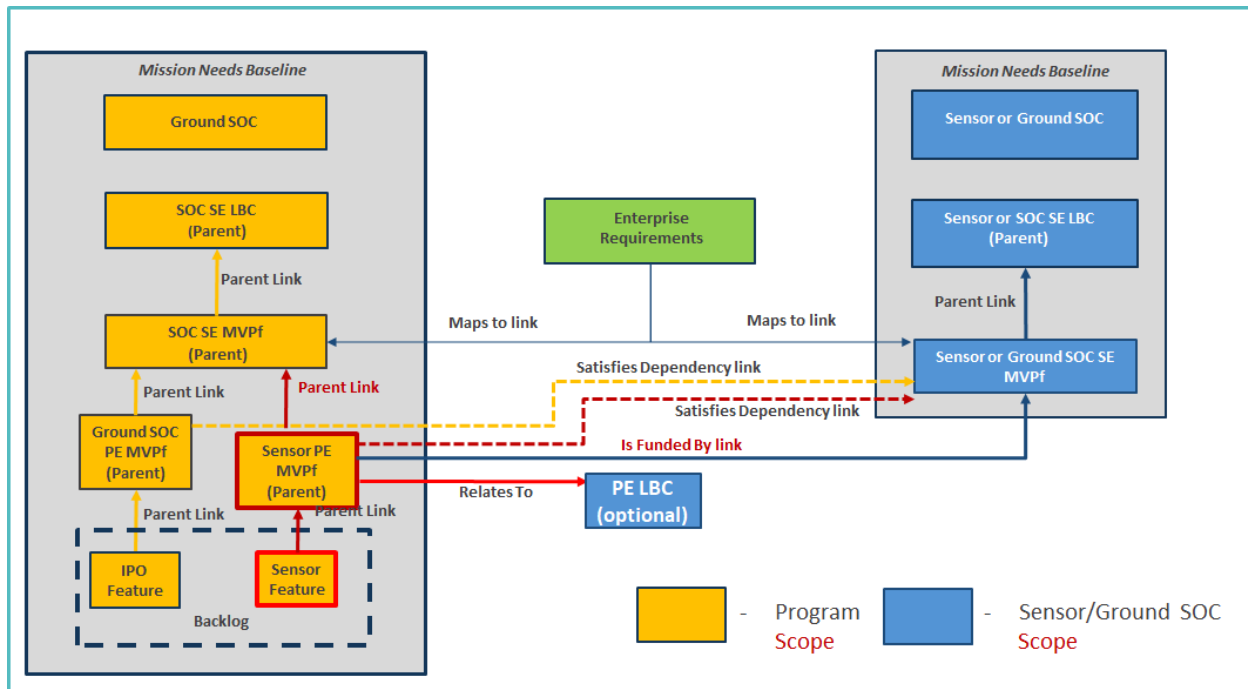
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Figure 3.1.5.1-1 (U) Requirements and Mission Needs Taxonomy**3.1.5.2 (U) Expanded Mission Needs Taxonomy**

(U) The basic Mission Needs Taxonomy does not define how to accommodate the addition of new requirements to the existing program baseline (e.g., when a Sensor Program Office allocates new requirements to another IPO or GPMO). To clarify this type of relationship in Jira, the basic Mission Needs taxonomy had to be expanded (Figure 3.1.5.2-1).



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Figure 3.1.5.2-1 (U) Expanded Mission Needs Taxonomy

(U) It is through the NGA RFC process that a Sensor Program Office, or another program, is able to allocate requirements and provide funding to another program office to conduct new, or additional development activities. It is recommended these requirements be negotiated between the requesting organization and program and that these requirements are captured in a PE LBC and included in the RFC. Once the RFC is approved, the implementing organization has formally accepted the new requirements, and they are now in scope. This enables the implementing organization to conduct new/additional development activities.

(U) Although the requesting Sensor Program Office or program is funding development activities and has stakeholder influence, it does not have responsibility or control of development activities. Funding is captured by addition of an “is funded by” link; the linkage is between the requesting organization SE MVPf and the funded Parent Organization PE MVPf. In some cases, the receiving organization may already be developing a capability that can be leveraged by the Sensor Program Office or program at no cost. In these instances, the “satisfies dependency” link is used.

(U) The use of the PE LBC remains optional, and if used in the context of the Mission Needs taxonomy the recommended PE LBC to PE MVPf relationship is captured using the “relates to” Jira issue linkage.

3.1.6 (U) Normalized Capability Roadmaps

(U) The tools used to create these normalized capability roadmaps are Enterprise Jira and Tableau®. Tableau extracts the SE, PE, and Feature data that is in Enterprise Jira to create Solution and Program-level Capability roadmaps. These Capability roadmaps are normalized across programs and are visible to the enterprise.

3.1.7 (U) Benefits

(U) The benefit of ERM is that it provides awareness of upcoming planning events and releases early enough to influence schedules and resolve potential conflicts. In order to achieve this, ERM monitors: schedules, synchronization/cadences, cross-program dependencies, strategic initiatives, and other cross-contract activities.

3.2 (U) Digital Engineering (DE)

(U) NGA employs a Model Based Systems Engineering (MBSE) methodology that focuses on creating and exploiting discipline¹-specific models as the primary means of information exchange, rather than a document-based information exchange.

(U) DE is a methodology that achieves a single comprehensive view of the system by integrating discipline-specific data sources, which are created using standard model elements (DoD DE Strategy Guide). In this context, the term “system” can be applied to technical (e.g., architecture, Information Technology (IT) infrastructure, development, test, security); financial (e.g., program & budget, cost-estimation); or process components (e.g., requirements, policy, configuration management, standards). DE enables integrated outcomes by creating the environment where authoritative system data from various sources can be consolidated using a common metadata format, which are analyzed with the aid of established relationships between the various data sources (Figure 3.2-1).

(U) NGA’s transition from legacy management process to a digital ecosystem that combines model-based techniques, digital practices, and an underlying computing infrastructure is being accelerated by employing DE, which is a variant of MBSE.

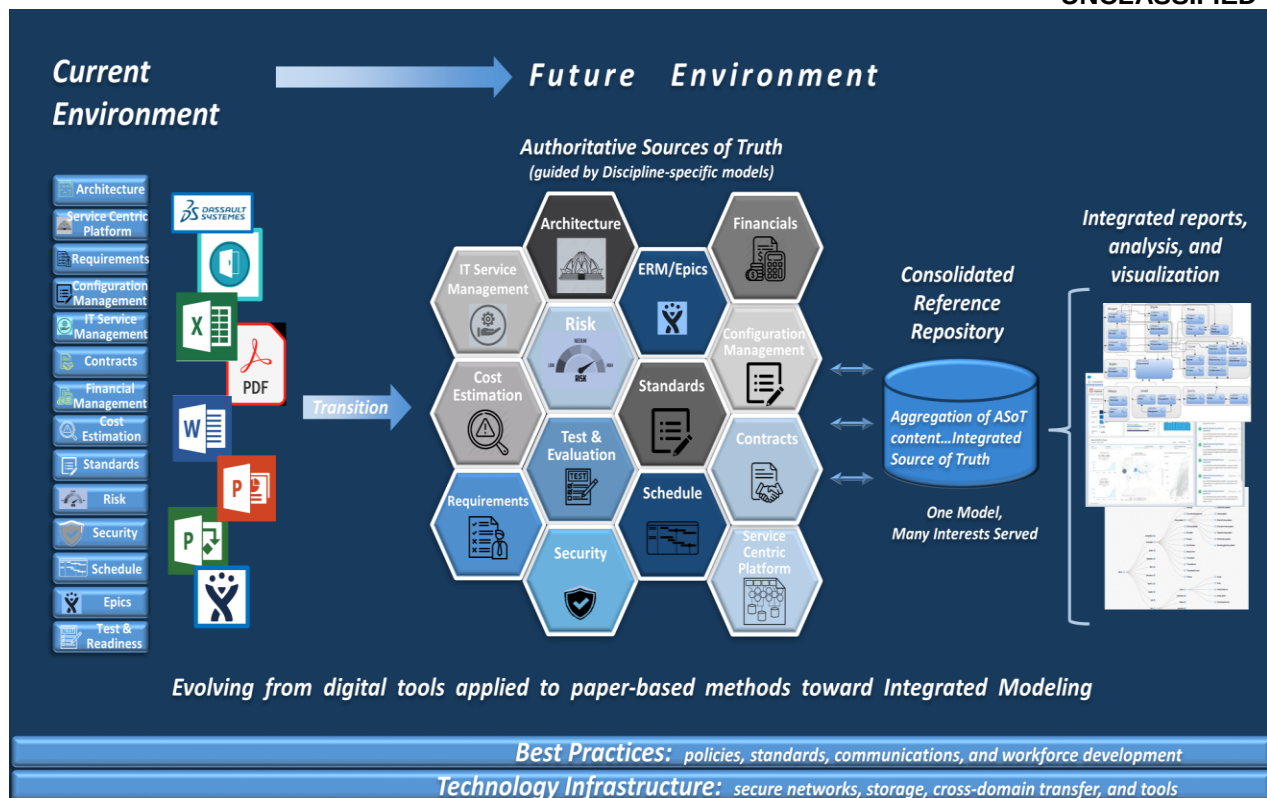
(U) The DE Ecosystem consists of the following:

- (U) Models which provide a digital representation of a specific system or operation and guide the creation and management of an Authoritative Source of Truth (ASoT);

¹ “Discipline” in this context refers to specific categories of information (e.g., requirements, architecture, project management/development, security, finance) that the enterprise uses for development & sustainment resource allocation decisions.

- (U) ASoTs are the datasets created by employing Discipline-specific models, that capture the current information state and provenance of DE baseline components;
 - (U) Consolidated Reference Repository (CRR), which is the data store that houses the unique data elements of all modeled information (i.e., ASoT) and establishes relationships between the discipline-specific ASoTs, and provides the data for integrated analysis and reporting;
 - (U) Best Practices that collectively are a group of policies, standards, communications, and workforce development which supports the adoption and execution of DE across the lifecycle;
 - (U) Technology Infrastructure that provides the secure networks, storage, cross-domain transfer, and tools that enable model creation and maintenance, data access, ASoT generation and use, relationship formation, and integrated analysis and reporting.
- (U) Working with stakeholders to create and maintain the DE ecosystem, improves transparency and traceability of requirements and budgets across the architecture from strategy to implementation, resulting in the more efficient and effective delivery of products and services.

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Figure 3.2-1 (U) NGA's Digital Engineering (DE) Ecosystem

(U) Additional details of NGA's DE constructs and implementation is described in the DE CONOPS.

3.3 (U) Architecture Engineering

(U) The primary objective of building and maintaining architectures is to provide an architecture ASoT for Digital Engineering.

3.3.1 (U) Levels of Architecture for NGA

(U) At NGA, the levels of architecture abstraction represent a continuum of architectural detail that enables tracking of Strategic Needs and Budget Program associations through assigned programs through SEs and PEs that define the deliveries of executable solutions over time.

3.3.1.1 (U) GEOINT Enterprise Architecture (GEA)

(U) The GEA encompasses existing and developmental, systems, services, and programs that support the mission of the Agency. The GEA is constructed with stakeholders, leveraging active data sets across the organization. GEA datasets are exploited through Architecture Frameworks to depict integrated views to describe the Agency Mission material and non-material capabilities and potential gaps. In migrating from a document-centric (e.g., Word, PowerPoint, and Excel) engineering content perspective to a DE data paradigm, with regard to architectural information, the enterprise is able to leverage models and data to analyze established relationships and export architecture viewpoints based on real time information. The GEA assists in guiding investments and acquisition activities to define the scope of development efforts, aid in integrating activities, and provide a common baseline for decision makers to guide efforts toward accomplishing mission success.

3.3.1.2 (U) Solution Architecture Supporting Community Acquisition Milestones

(U) Programs defined through Intelligence Community (IC) and Department of Defense (DoD) capabilities acquisition must conform their architecture description for Independent Cost Baseline Document (ICBD), the IC Program Architecture Guide (IC PAG) and/or similar DoD instructions. Programs may create architectural artifacts as needed to support other engineering activities such as requirements decomposition, CONOPS development, and risk mitigation

3.3.1.3 (U) Solution Architecture Supporting Solution Epics

(U) SEs decompose program solution architectures into relatively self-contained modular pieces for development planning. SEs define criteria that drive solution architecture models and associated requirements suitable for agile prioritization and even further decomposition. This decomposition of development work into SEs allows

programs to coordinate, leverage, and de-conflict development work and dependencies across the program functional scopes.

3.3.1.4 (U) Program Architectures Supporting Agile Implementation

(U) PEs are the result of decomposition of solution criteria into self-contained modular pieces for program development planning. PEs define the criteria that drive program architecture models and associated requirements/features suitable for agile prioritization and allocation to development teams. Breaking development work into PEs also allows the programs to coordinate, leverage, and de-conflict development work across multiple development contracts or agile teams.

3.3.2 (U) Architecture Governance

(U) Architecture governance is used protect the integrity of accepted architecture at the Strategic, Enterprise, and System/Service levels; and to maintain its overall coherence in the face of change. NGA's architecture governance approach is to modulate the acceptance of architectural content using decision gates at key points of entry into the repository. These decision gates represent a disaggregation of authority down to the lowest levels where proposed and planned design decisions are made. By sharing a common repository, standards, instructions, and tools, all architecture content is shared up and down the architecture layers between the GEA and Program Management Offices (PMO) developing solution architectures representing conceptual, proposed, planned, and operational designs.

3.3.3 (U) Architecture References

(U) The following artifacts are maintained to promote architecture development at NGA:

3.3.3.1 (U) GEA Website

(U) The GEA website presents GEA architecture information to all NSG users. The GEA website includes the following tabs:

- GEA Landing Page: Contains a simplified DE metamodel linking across GEA Website
- GEA Navigator: Allows non-architects to investigate GEA relationships
- GEA Baseline: Contains all current GEA Baselined products
- GEA Change Log: Tracks historical changes to the GEA
- GEA Visualizations: Provides additional visualizations to investigate GEA relationships
- IC Joint Architecture Reference Model (JARM) / Technology Services Taxonomy (TST) 5+: Provides access to IC JARM documentation
- Architecture Practitioner Page: Provides access to architect resources

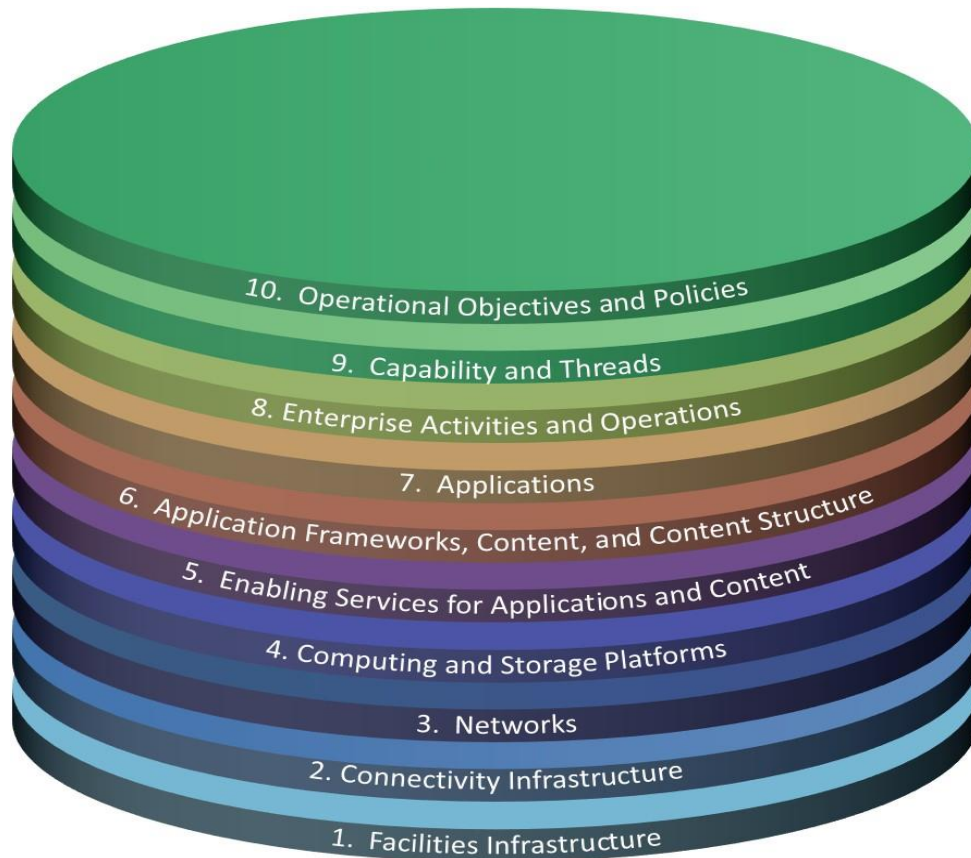
- Architecture related Policy: Lists all applicable architecture policies and presentations

3.3.3.2 (U) IC Joint Architecture Reference Model (JARM)

(U) The IC JARM is an extension to the Federal Enterprise Architecture (FEA), tailored to the activities, functions, and services unique to the IC. The JARM consists of three interrelated taxonomies overlaid on a single organizing construct, the JARM 10-Layer Model (Figure 3.3.3.2-1).

(U) The model is typically discussed from the top (Layer 10) to the bottom (Layer 1) and each layer of the model represents a logical grouping of like entities, capabilities, functions, or services within an enterprise. The JARM supports architectural analysis and reporting, increased communication, reusability, and interoperability across IC Mission Elements. NGA captures desired updates and changes to the IC JARM Technical Service Taxonomy (TST) in a Microsoft Excel file called the IC JARM TST 5+.

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Figure 3.3.3.2-1 (U) JARM 10-Layer Model

3.4 (U) Configuration Management (CM)

(U) CM is an essential element of systems engineering which provides the mechanisms and procedures for maintaining product consistency and establishes an appropriate level of configuration control for NGA's evolving systems. CM ensures that: interfaces are defined, designs trace to requirements, documentation is accurate, and change is documented and controlled. The NGA Configuration Management Plan (NCMP) details NGA's configuration management procedures.

3.4.1 (U) Configuration Baselines

(U) The NCMP categorizes the NGA technical baselines as Functional, Allocated, and Product. In general terms, these baselines are described as follows:

- **Functional Baseline:** defines functional, performance, interface, interoperability, and verification requirements. It is established with the approval of the top-level requirements documents (e.g., RD, DD, and COR). The NGA CM libraries are the repository for the functional baseline. Typically, the functional baseline is established during the pre-acquisition phase of a project and managed through completion of a project or disposal of a system.
- **Allocated Baseline:** defines design and implementation requirements. It is composed of the design documentation as well as the hardware and software items being developed. An allocated baseline is established during the design phase of a project, and managed through the end of development/integration.
- **Product Baseline:** describes the as-built system; and user, operations, and support requirements. It is composed of the hardware and software items of the operational systems. The Product Baseline is established as the project transitions to operations, and is managed through system disposal.

3.4.2 (U) The CM process includes five main components

- CM Planning – The identification of the processes and procedures used to manage the configuration of systems and the changes to those systems throughout the life cycle.
- Configuration Identification – Identifies and documents the functional and physical characteristics of configuration items and establishes appropriate baselines.
- Configuration Control – Evaluates, coordinates, and approves proposed changes to an established baseline.
- Configuration Status Accounting – Records and reports the status of configuration items and change requests (proposed or approved).
- Configuration Audits – Periodically review all controlled items to ensure that they evolve properly with recorded traceability and that they meet program needs.

3.4.3 (U) Configuration Control Authority

(U) The NGA CCB serves as the final approval authority for technical/functional baseline changes for IT. Configuration control of the development and operational baseline has been delegated to the programs and the NGA Chief Information Officer and IT Services (CIO-T) operations group. The Configuration Items (CIs) controlled by the CCB are hosted in the Baseline Management System (BMS).

(U) The NCMP describes NGA CM-specific information such as NGA Configuration Control Authorities (CCAs), the Engineering Review Board (ERB), IT governance hierarchies and relationships, change control and implementation procedures, and Enterprise Requirements Hierarchy. In addition, the Tier 2 CM Procedure is available

and addresses detailed guidance for RFC; Discrepancy Reports and Problem Reports; Change Implementation; Verification and Audits; Documentation Preparation and Maintenance; and Schedule Management.

3.5 (U) Readiness and Technical Reviews

(U) NGA supports planning for NSG milestone review and associated activities in accordance with the Tier 2 NGA Enterprise Readiness (ER) Process Document. The purpose of the NGA ER process is to facilitate efficient and effective Acquisition Readiness reviews that assess whether segments, systems, services, capabilities, and/or applications are ready to be moved through the identified milestones/reviews in the NGA Acquisition Lifecycle; including into or out of the operational environment. The NGA ER Process is designed to accommodate the need for flexibility, by tailoring to the specific needs of the acquisition effort. The process documentation includes readiness checklists to help facilitate planning for and execution of technical review activities. The readiness process also addresses programs that are utilizing agile development and provides a process for evaluating readiness to operate in the Cloud. Agile development checklists are available for Release Planning Events (RPEs) and Final RPEs (FRPEs) and located on the Integration Test and Image Quality Office (VOLT) website.

(U) Within the IC, the Office of the Director of National Intelligence (ODNI) holds the Milestone Decision Authority (MDA) for all MSA Milestone A and B events. If and when the ODNI delegates the MDA authority to NGA for Milestone A and B events, the Component Acquisition Executive (CAE) will serve as the MDA. The MDA for Milestone A and B events cannot be delegated below the CAE. The NGA CAE for all other non-MSA reviews may delegate the decision authority responsibility for Level-0 and Level-1 Readiness Reviews.

(U) NGA's Assessments Division within the Architecture and Engineering Office works in coordination with the appropriate Office of Primary Responsibility (OPR) or Directorate and identifies a primary and alternate MDA for all milestone readiness events and reviews.

(U) The NGA ER process is flexible and can be tailored to the specific needs of the acquisition effort. The ER Process can support programs that employ either traditional or agile engineering processes. The NGA ER process supports acquisition efforts that require significant technical evaluation, such as an MSA. The process also supports other acquisition efforts of small scope, low risk, and proven technology or requiring quick response in support of crises, justifies tailoring the readiness process.

3.6 (U) Requirements Engineering

(U) The approach to engineering requirements followed by the NGA is documented fully in the CCB approved document, NGA Enterprise Requirements Process (NERP). The NERP identifies relationships with other engineering disciplines identified in the SEP and illustrates the dynamics with each in order to provide an actionable and defensible

requirements baseline. Roles of the stakeholders are identified in context of requirements engineering in the NERP. NGA's process for enterprise requirements follows the same major activity paths, whether an executing budgeted program office is working a traditional or agile path. The Standard Operating Procedures (SOP) associated with each layer of the NGA requirements process are provided in the NERP.

(U) NERP Goals:

- Identify, document, and maintain requirement baselines
- Manage and control requirement baselines using approved configuration control methods
- Document horizontal and vertical requirement traceability
- Align requirements to architecture

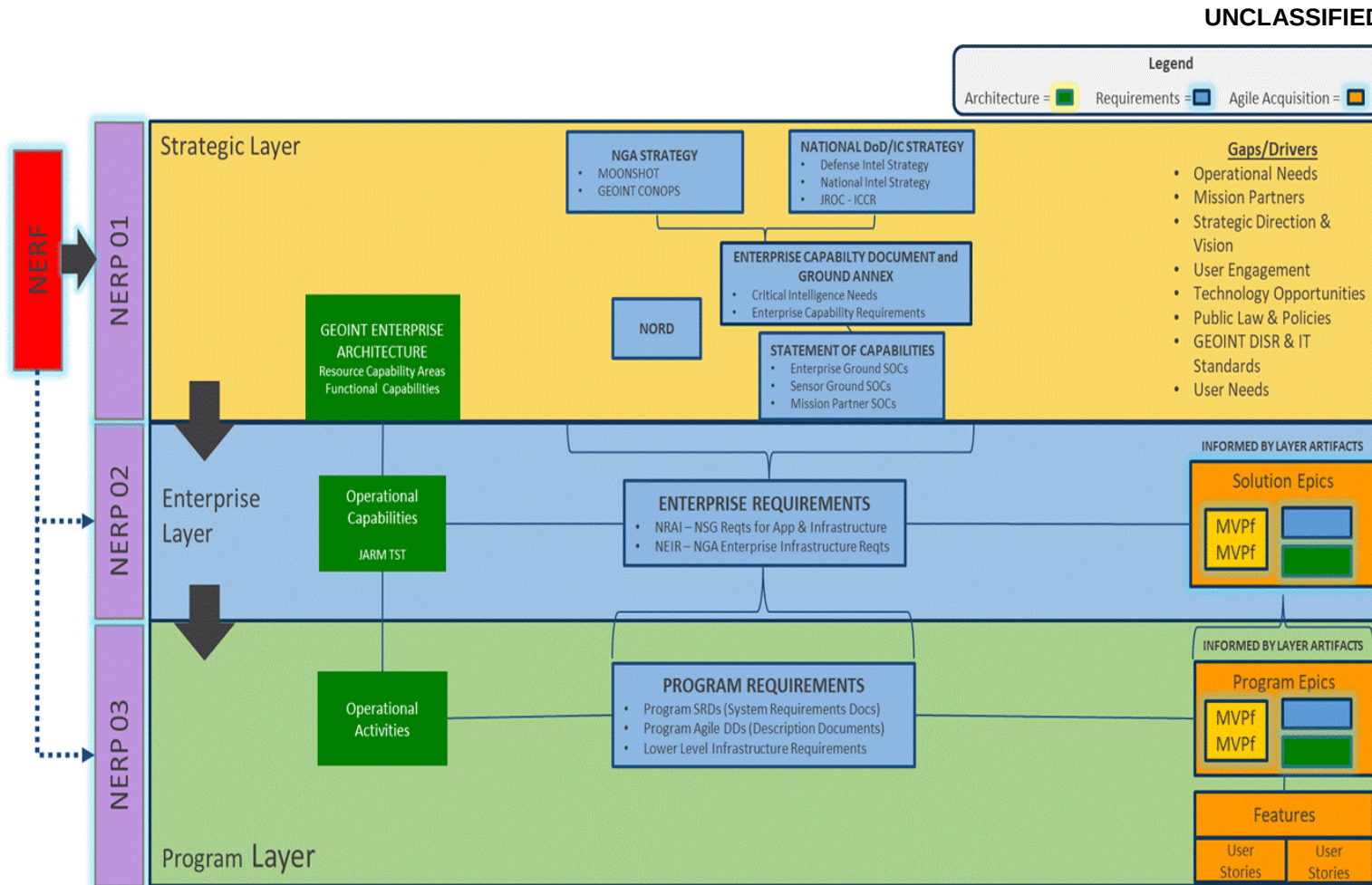
(U) The NERP has three sub-processes (Figure 3.6-1) that address the varying levels (layers) of requirements at NGA. The sub-process NERP 01 governs activities occurring within the Strategic Layer, NERP 02 governs the activities within the Enterprise layer, and NERP 03 covers the Program layer

(U) It is possible to tailor individual layers within the NERP. For the purposes of this document, tailoring is defined as "the adaptation of a process for a particular purpose or individual customer" while fulfilling NERP goals.

(U) The NERP connects to and results from the greater NSG Enterprise Request Framework (NERF), managed by NGA's Partner Engagement Integration Division. The NERF adjudicates a request from mission partners against baselined requirements to align with known scope for budgeted programs when possible via NERP. Where scope and/or requirements do not exist, the NERF activities flow up to NGA leadership for decision. The NERF adjudicates a request from mission partners against baselined requirements to align with known scope for budgeted programs when possible via NERP.

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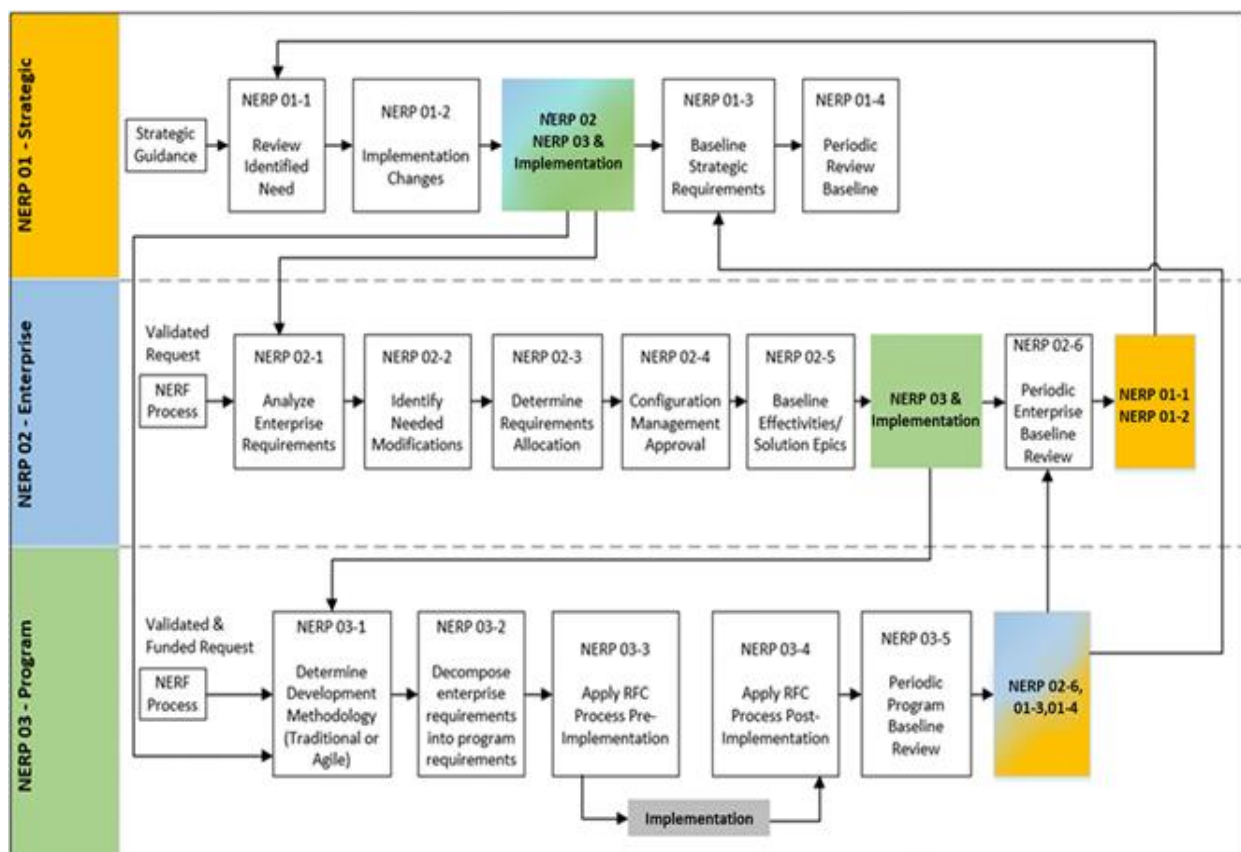
Figure 3.6-1 (U) NERP Processes and Artifacts by Requirements Layer

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(U) The NERP processes provide the following:

- Perform needs and requirement analysis to validate needs; establish baselines and allocate requirements; ensure baseline traceability; conduct analysis of alternatives and identify courses of action; and ensure baseline integration and normalization (Figure 3.6-1).
- Perform requirement analysis to validate needs, establish baselines, and allocate requirements that support national strategy and NGA strategic objectives and guidance.
- Ensure requirements management through CM controlled governance processes and procedures.
- Support program analysis to interpret strategic and enterprise requirements for implementation.

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Figure 3.6-2 (U) NERP Processes

3.7 (U) Risk, Issue, and Opportunity Management

(U) Risk, Issue, and Opportunity (RIO) management is the identification, assessment, and prioritization of RIOs followed by the coordinated and economical application of resources to minimize, monitor, and control the probability and/or impact of unfortunate events or to maximize the realization of opportunities. Risk management's objective is to assure uncertainty does not deflect the endeavor from the project goals.

(U) NGA utilizes a tiered approach built on DoD and industry standards to manage RIOs, (Figure 3.7-1), as further detailed in the Information Technology Risk Management Process (ITRMP). The responsibilities for managing risks are shared throughout the agency from the highest levels of executive leadership to managers executing program and project activities. This layered approach empowers management at various levels to develop right-sized plans, processes, procedures, and tools with the goal of managing RIOs at the lowest possible level. Each level should include an escalation process to the next higher level.

(U) Tier 1 risks and risk programs include input and efforts from program management offices (PMOs) and project managers, and is the primary source for RIO identification. Program risk management plans include a workflow that illustrates how a Program Management Office (PMO) handles program-level, or Tier 1, RIOs, and provides details on steps required to nominate RIOs for Tier 2 escalation.

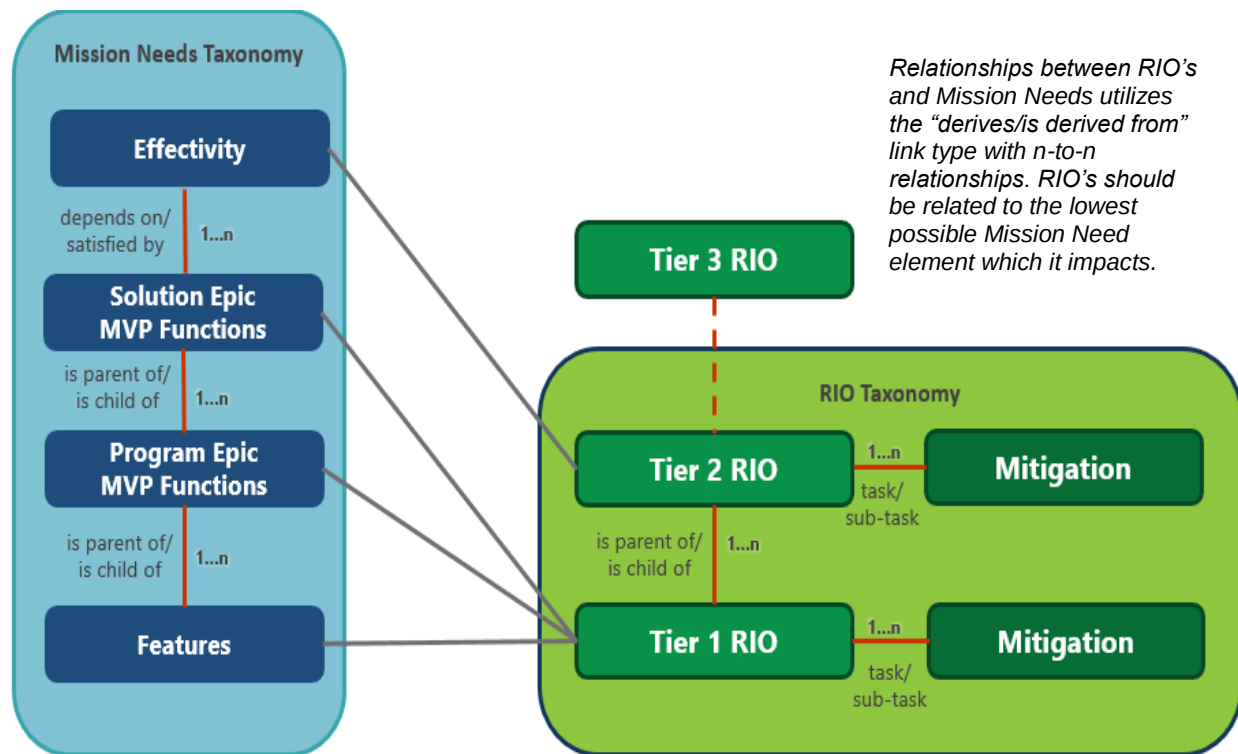
(U) Tier 2 RIOs impact multiple PMO's, Mission Partners, or are outside the control of the Program Manager. RIOs nominated for Tier 2 management will be adjudicated following the IT Risk Management Process as documented in the baselined IT Risk Management Plan (RFC: N01-0709). Should a RIO require Tier-3 Strategic coordination, appropriate escalations paths, as outlined in the IT Risk Management Plan, will be followed.



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Figure 3.7-1 (U) RIO Tiers

(U) In alignment with NGA's DE environment, Tier 1, 2, and 3 RIOs are managed and tracked per ERM tools and procedures, leveraging Jira as the primary tool for managing risk. Within Jira, program offices have projects built into the Risk Management schema, or template, that all Tier 1, 2, and 3 Risks use. The schema provides a consolidated risk management environment to ensure all program offices use the same fields and field definitions. Specific linkage relationships and business rules are documented in the DE Metamodel. Linkages between the RIO taxonomy and the ERM Mission Needs taxonomy is shown below in [Figure 3.7-2](#).



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Figure 3.7-2 (U) RIO Jira Taxonomy

3.8 (U) Schedule Management

(U) As an IT Enterprise, NGA's schedule management processes, procedures and practices are designed to ensure a well-defined, standard schedule hierarchy, business rhythm and accepted best practices. These provide consistent information sharing across the agency. NGA has schedule managers at the enterprise, office, and program levels. The office and program schedule managers are responsible for developing and/or validating schedules and providing the appropriate schedule information to be captured at the enterprise level. The enterprise schedule managers are responsible for the management and integration of schedule activities and dependencies in enterprise schedules including the NGA Integrated Schedule (NIS), and the NGA Configuration Control Board (NCCB) Managed Schedule (NMS).

(U) The enterprise has a defined collection of capability deliveries, known as Effectivities. These effectivities allow for the temporal collection, integration, testing, and deployment of operational capabilities to address user needs. The content of each effectivity is governed via the RFC process and controlled by the CCB to ensure coordination and approval of deliveries. Dates associated with each effectivity are

baselined as part of the NMS. Effectivities can also be used to define program specific deliveries.

(U) At NGA, agile and traditional methodologies are employed for program planning/development and their related schedules. The schedule processes defined in the NGA Schedule Management Plan (NSMP) apply regardless of the methodology used. The NSMP provides flexibility for programs and projects to govern their own scope and schedule, as defined by their engineering processes.

(U) The enterprise schedule process integrates program and enterprise schedules, and aligns with the IT Program Objective Memorandum (ITPOM), GEA, and NSG strategic plan as well as planned and funded services. The process incorporates a tiered hierarchy, where program-level activities and milestones support higher-level effectivities or new capabilities. Offices and programs nominate top milestones for reporting in the NIS. The schedule management process establishes the foundation for management of lower-level schedules, according to commercial best practices, collection of key milestones and cross-effort dependencies, and reporting of impacts to appropriate levels of leadership and mission partners. The contents and schedules are hosted as defined in the NSMP; this allows projects and programs to manage schedules within the enterprise context and facilitates rigorous enterprise-level engineering analysis (such as critical path analysis). Specifically, schedules are maintained via a combination of Jira project and MS Project files and visualized using Tableau.

(U) Tableau schedule visualizations provide near-real time depictions of the enterprise schedule data. Schedule depictions are primarily Gantt formatted information filtered for Enterprise activities, with drill down capabilities to the lower level detailed schedule data. Specific data sets are incorporated based on source data availability and the purpose of the visualization (e.g. Executive Overview, Moonshot Overview, Test Planning, etc.). Visualizations created from Jira data sources are automatically updated on a nightly basis.

3.8.1 (U) Effectivities and the Mission Needs Taxonomy

(U) The Enterprise Schedule Management (ESM) team incorporates Effectivities in the NIS to capture the dates when new or modified capabilities are going to be added to the current baseline. An Effectivity is dependent on the completion of project/program development activities by the Effectivity's specified due date.

(U) The ESM team has established a taxonomy that links Jira-based development activities to Effectivities found in the Jira-based Effectivities Project. The Jira-based Effectivities project is a duplicate of the Baseline Management System (BMS) Effectivity Management log, which is the ASoT for Effectivities. The Jira Effectivities project receives near real-time Effectivity updates.

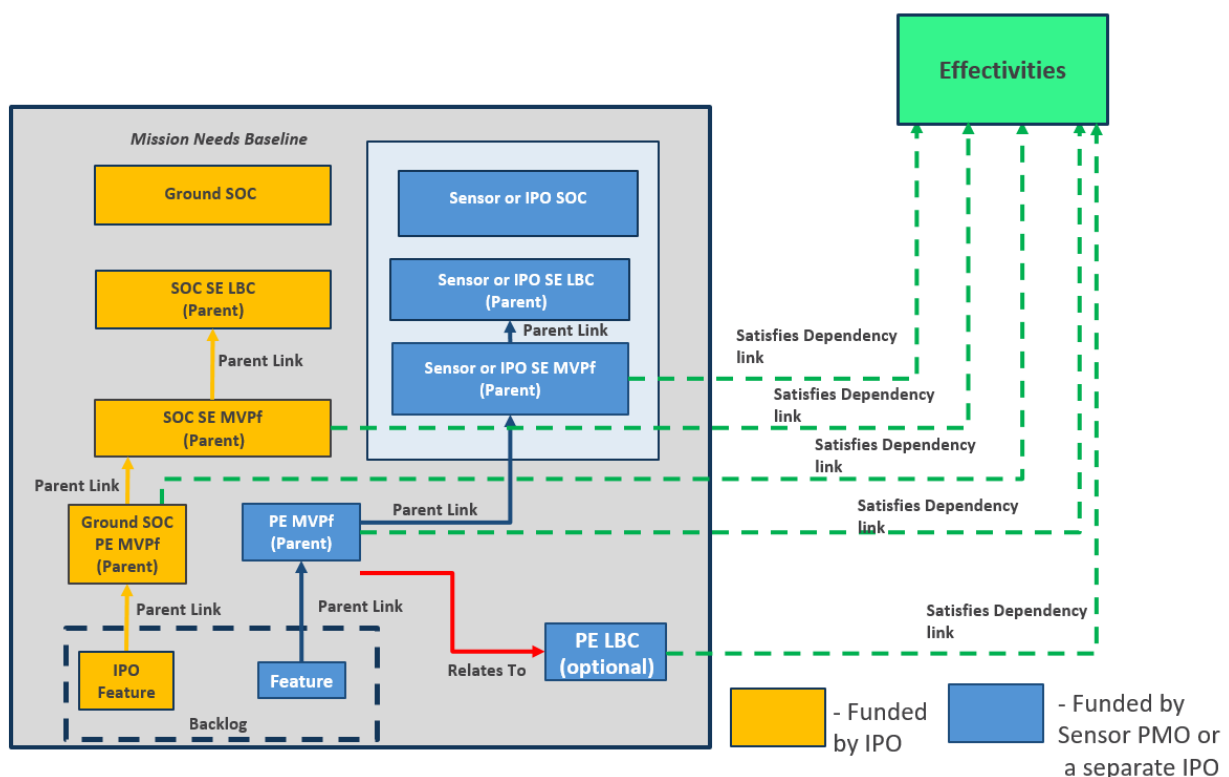
Figure 3.8.1-1 shows the relationship between the Jira-based Effectivities Project and program office Jira projects. The relationship can be established at either the

Enterprise, or Program levels. Either an SE MVPf, PE MVPf, or PE LBC can satisfy an Effectivity, but not all.

- If all PE MVPfs in an SE MVPf are required for an Effectivity, then the linkage is made to the SE MVPf
- If some of the PE MVPfs are not required to satisfy the Effectivity, then only the relevant PE MVPfs are linked
- If a PE LBC is used to collect all relevant Effectivity PE MVPfs to be developed by a single contractor, then the linkage will be established to the PE LBC.

(U) The Effectivity is dependent on development activities described in the SE MVPf, or PE MVPf, or those captured in a PE LBC. The linkage used in Jira is a “Satisfies Dependency” link.

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Figure 3.8.1-1 (U) Effectivity to Mission Needs Taxonomy Relationship

3.9 (U) Security Engineering

(U) NGA programs are responsible for the integration of GEOINT services and capabilities into the NSG enterprise. Those programs may include National Security Systems (NSSs) that process classified information concerning or derived from intelligence sources, methods, or analytical processes, which is required to be handled within formal access control systems established by the Director of National Intelligence. The Committee on National Security Systems (CNSS) Instruction No. 1253, "Security Categorization and Control Selection for National Security Systems," provides all Federal Government departments, agencies, bureaus, and offices with processes for security categorization of NSS that collect, generate, process, store, display, transmit, or receive National Security Information.

3.9.1 (U) Security Engineering Objectives

(U) From a security integration and oversight perspective, and using the National Institute of Standards and Technology (NIST) Risk Management Framework, the program office must ensure that each affected segment understands its security requirements, works to build in the security controls in the Design phase, and is marching to the same schedule for test and authorization. The schedule is developed using CNSSI 1253 security controls compliance as primary consideration. The initial baselined controls are determined by the overall impact level of information types, identified to determine the highest impact level (low, moderate, or high) for each information type. Each of the affected program segments will maintain their Security Plan Identification (SPID), which documents the program's compliance with identified objectives and oversight mandates.

3.9.2 (U) Security Engineering Approach

(U) Information Assurance (IA) is addressed throughout the system life cycle in accordance with ICD 503 and CNSSI 1253. IA protection considerations occur throughout the System Development Life Cycle (SDLC) with security built-in at the program's inception. This enhances overall system security, reduces costs, and shortens the authorization process. The NIST Risk Management Framework provides a comprehensive governance structure and organization-wide risk management strategy to be employed throughout the SDLC and includes the roles and responsibilities required to effectively conduct IA. Risk management tasks must begin early in the SDLC and are important in shaping the security capabilities of NGA Enterprise.

(U) If a program vendor does any application development or modification, the vendor then has the responsibility to scan and mitigate application security vulnerabilities prior to deployment. If the NGA program drives architecture/configuration changes to an existing NGA System/Service/Segment/Entity, then upon each release the program submits a Security Impact Decision Request (SIDR) form for review. If the change is

considered security-relevant, then the system must go back through the risk management framework process to be evaluated for additional security controls.

3.9.3 (U) Security Engineering Software Approval

(U) NGA Programs use only applications listed on the NGA Whitelist. These applications are authorized for use on any NGA system, as long as the software is still vendor-supported.

(U) For any new software to be added to the NGA Whitelist, it must undergo the Software Research and Review process and approval by the Software Approval Process (SWAP) team. All requests are submitted via the online SWAP tool. This tool is used to request the addition of new software to the NGA Whitelist, as well as version upgrades of existing SWAP approved software items. The SWAP process is initiated by completing an online form within the tool. Once this form is completed and submitted the program Point of Contact (POC) will be sent notification by the SWAP team to review the request. The Program POC will review the submission for accuracy of program needs, and ensure a valid software version, and Uniform Resource Locator (URL) have been provided for SWAP team verification. The Program POC receives either approval or disapproval of the software whitelist request from the NGA Test and Image Quality Office SWAP Team.

(U) The SWAP strengthens the security of NGA systems and networks by ensuring that only reviewed and approved software is installed and used. Programs wishing to use a particular application must follow the NGA process. This applies to open source, freeware, shareware, Commercial Off-the-Shelf (COTS), and Government Off-the-Shelf (GOTS) applications.

3.10 (U) Standards Management

(U) Standards adherence enables interoperability and net-centric services across the IC, DoD, and the US Government. Along with requirements, architecture, and capability definitions, standards serve to guide acquisition and development activities. Standards compliance promotes consistency of services and attributes, which enable interoperability, performance, cost effectiveness, maintainability, and security. There is a wide range of applicable standards that engineering activities must support and apply for conformance and compliance in data, services, systems and activities. NGA services (including NSG services) required to meet NGA Functional Management responsibilities for use by DoD, must be certified as conformant with prescribed DoD standards. Therefore, the definition, maintenance, and application of a variety of standards are intrinsic to the NGA engineering process.

(U) The Standards Division is responsible for representing NGA within the joint DoD and IC Enterprise Standards Baseline (ESB) process, leading one of the process's technical working groups, the GEOINT Working Group (GWG). The GWG is also the U.S. Government's NSG forum which leads and manages GEOINT standardization activities

to meet the mission needs and strategic direction of the DoD Joint Information Environment (JIE) and IC IT Enterprise (IC ITE). The GWG advances GEOINT interoperability at three levels: geospatial data exchange, network services, and systems.

(U) In addition to the responsibilities for GEOINT standards, the Standards Division represents NGA in the joint DoD and IC standards adoption process for all other IT standards (e.g. IEEE, IETF, W3C, ANSI, and ISO). Applicable IT Standards, including GEOINT standards, are officially cited in the DoD IT Standards Repository (DISR) and are designated for applicability and compliance, as follows:

- **Mandated:** these standards provide interoperability and net-centric services across the DoD and IC enterprise. They are the minimum set of essential standards for the acquisition of all DoD and IC systems that produce, use, or exchange information and, when implemented, facilitate the flow of information in support of the warfighter. These standards are required for the management, development, and acquisition of new or improved systems throughout the DoD and IC.
- **Mandated Sunset:** these standards have been identified and approved through the joint DoD and IC enterprise standards baseline process to be tagged for retirement. A pre-defined event and date are associated with the standard sunset tag. In addition, a replacement standard is frequently identified. In the DISR online an “X” marking in the “sunset” column in a standards profile identifies this sunset status. The sunset tag does not require a waiver or a migration plan.
- **Emerging:** these standards may be implemented, but shall not be used in lieu of a mandated standard but may be used in addition to a mandated standard. An emerging standard is expected to be elevated to mandatory status within three years. Use of an emerging standard in a Technical Standards View Profile (StdV-1) requires a waiver and a Technology Insertion Risk Assessment. In general, emerging standards should be placed in the Technical Standards View Forecast (StdV-2).
- **Information/Guidance:** these are IT-related documents developed by a process outside the standards-development process (e.g., DoD Directives, Instructions, handbooks, manuals, procedures, best practices, and other such policy documents) that offer a means to further clarify standards and identify relevant policies and procedures. Such documents are not mandated for use for net-centric interoperability.

(U) As an aid to acquisitions, the Standards Division maintains the Standards Selection Tool – Fit for Purpose (FfP). The FfP tool provides program offices, program managers, engineers, contractors and acquisition activities an improved method to identify and access standards aligned with the Intelligence Community (IC) Joint Architecture Reference Model (JARM). To maximize efficiencies and reduce costs, use of the FfP

standards tool is necessary in the development of standards technical views for the IC and DoD.

3.10.1 (U) GEOINT Standards

(U) GEOINT Standards consist of technical specifications or other precise criteria to be used consistently as rules, guidelines, or definitions of characteristics of GEOINT to ensure that materials, products, processes, or services are fit for the analysis and visual representations of physical features and geographically referenced activities on, in, or near the Earth.

(U) The Standards Division leads and administers the GWG, the forum that manages the adoption and citation of GEOINT standards for the DoD/IC. In this role, the Standards Division actively monitors and participates in the GEOINT standards development and endorsement community. In the absence of an open source, voluntary consensus standard, the Standards Division, through the NGA Functional Management authorities, works with Standards Development Organizations to influence their development process to address NSG needs; develops and publishes profiles of such standards where possible; or develops an NSG standard.

(U) The main program for conformance assessment of GEOINT standards is the GEOINT Functional Manager Standards Assessment (GFMSA). The GFMSA program defines and implements NSG methods and processes to assess and assert IT conformance to applicable GEOINT data and service standards.

(U) GEOINT standards are accessible via the NSG) Standards Registry. The NSG Standards Registry supports the discovery, traceability, and lifecycle management of GEOINT Standards that may be used in the development and operation of data and net-centric GEOINT applications.

(U) In addition to standards, NGA relies on the documents and engineering artifacts associated with the NSG Application Schema (NAS), which specifies an NSG-wide logical model for geospatial data that is technology neutral. This Platform Independent Model determines the syntactic structure used to represent the semantics specified by the NSG Entity Catalog (NEC). From it, using Model Driven Architecture (MDA) techniques, technology-tied Platform Specific Models (PSM) may be automatically derived and directly employed in system development.

3.10.2 (U) IT Standards

(U) The Standards Division represents NGA in the joint DoD and IC ESB process, working with the process's other technical working groups to identify, adopt, and manage the citations for all other IT standards that are also cited within the DISR. The Standards Division provides NGA visibility of proposed ESB changes, entertains comments on adoption recommendations, and represents NGA in the final adoption phase of the joint DoD and IC ESB process.

(U) The Standards Division also participates with the DoD's Defense Standardization Program (DSP), as NGA's Deputy Standards Officer (DepSO), in the review and publication of non-IT defense and federal specifications and standards, military handbooks, commercial item descriptions, data item descriptions, and related technical documents, through the DSP's ASSIST website.

(U) In the international arena, the Standards Division participates in the Defense Geospatial Information Working Group (DGIWG); the NATO Standards Agreement (STANAG) process, including the custodianship of key GEOINT standards (STANAGs); the International Organization for Standardization (ISO) Technical Committee (TC) 211 on Geographic Information/Geomatics; the Open Geospatial Consortium (OGC) and others, to influence the development of standards to address DoD/IC/NSG needs.

3.11 (U) Test and Evaluation

(U) The NGA NSG/ASG Enterprise Test and Evaluation Master Plan (TEMP) describes the life-cycle independent Test and Evaluation (T&E) activities required to support the integration of hardware and software baselines across the NGA and user communities. The purpose of the Enterprise TEMP is to provide an overarching master plan for independent T&E of all changes to the NSG/ASG Enterprise prior to their operational deployment, specific segment/MSA capabilities T&E programs are not highlighted in the NSG/ASG Enterprise TEMP but rather are described in individual program addendums (program specific T&E tailoring plans) to the NSG/ASG Enterprise TEMP (as completed by the MSA). The NSG/ASG Enterprise ties together all parts of the NGA organization at all security levels, including corporate applications, desktops, network infrastructures, enterprise servers, Geospatial Intelligence systems, and the systems that NGA develops and delivers to its Community customers. Continual testing of the NSG/ASG's constituent systems is required to keep the operational environment working as efficiently as possible and to minimize disruptions while introducing new and changed capabilities to the NSG/ASG baseline. The TAC Organization is responsible for the NGA T&E process and for independent testing of both internal and external NGA deliveries, as such all NSG/ASG Enterprise TEMP addendums should be coordinated and approved by TAC. Systems engineers on projects and programs are responsible for ensuring that their segments or other entities go through formal developmental, integration, technical and operational evaluations, including independent verification and validation (IV&V) testing to ensure error free integration and operation as specified in the developer's program-specific test and evaluation management plan and/or as required NGA NSG/ASG Enterprise TEMP addendum, prior to deployment.

3.12 (U) Additional Systems Engineering Elements

3.12.1 (U) Agile Acquisition

(U) Traditional NGA acquisition is oriented toward large programs that produce significant changes in NGA capability over an extended period. In contrast, agile

acquisition utilizes agile methodologies and practices that allow greater flexibility to deliver the end users mission needs. These agile methods and practices allow the near term delivery of value to the end user. Agile places emphasis on the customer, which is accomplished through the early and continuous delivery of value by welcoming changing requirements and validation of software by the end user. This agility is accomplished by defining smaller portions of capability and delivering each portion incrementally in a shorter period, so users see small but frequent increases in capability. In this way, users directly influence the evolution of capabilities. In agile, working software is the primary measure of progress. While Agile acquisition has been a goal for some time, NGA is placing an increasing focus on agile responsive acquisition for the NSG program. Details regarding agile practices and agile acquisition can also be found in the ERM Confluence Site. There are many other excellent sources for agile methods and practices such as the Agile Government Leaders web site, 18F web site, NGA 5000.1 “Agile Acquisition,” and the United States Digital Services web site.

3.12.2 (U) DevSecOps Development Approach

(U) NGA is embracing the benefits of DevSecOps and is on its journey into the realization of a fully implemented DevSecOps Ecosystem. In this context, the term DevSecOps is inclusive of any prerequisite steps associated with delivering software to the production environment – to include security testing/approval, quality assurance testing and approval, and any other reviews or approvals needed to support the “go live” decision. It is essential that these intermediate steps have automated or semi-automated decision-making. While specific DevSecOps practices may vary across teams, there are fundamental characteristics that they all share:

- 1) (U) Tooling - There is a suite of tooling that facilitates the continuous nature of DevSecOps from development to deployment into the target environment;
- 2) (U) Comprehensive Automation - Automation of documentation, inspection, verifications, and validations to ensure fast and consistent deployments of products to operations; and
- 3) (U) Culture - The culture of the development team, product management team, and oversight teams (e.g. NTO, Security, etc.) and operational support teams embraces DevSecOps practices (i.e. change the way they view the Enterprise and product lifecycle).

(U) DevSecOps’ Continuous Deployment to the production environment is not necessarily the end goal, and must be weighed against mission and operational considerations; however, Continuous Deployments to a test/stage environment should be a goal, as that practice helps to quickly identify a variety of different potential issues, which when identified earlier in the development lifecycle will reduce capability costs and ensure business value is continuously delivered to customers.

(U) DevSecOps is a multifaceted discipline that spreads across typical IT organizational verticals; no longer are there separate development teams, security teams, test teams, and operations teams, there is only one team, the product team. It is the responsibility of the product team to ensure the product's lifecycle success to include rapidly fixing security and functional issues that emerge in operations. This is not saying the vertical organizations have to change their organizational structure, but it is saying the vertical organizations have to execute their business areas differently; more specifically be an active part of a product's success versus be a potential blocker.

(U) NGA currently maintains a platform of DevSecOps tooling for developers. It is in NGA's best interest for developers to leverage the existing DevSecOps tooling and example pipelines vice building their own from scratch. Developers are encouraged to offer their contributions to example pipelines, so that all NGA Developers may take advantage of each other's expertise and experiences. The only exception to this is when the current tool offerings do not meet development or operational needs. When this occurs a request for the required tooling needs to be made, and if the tooling cannot be provided; then the building of necessary tooling is permitted. Posting documentation of the build process to a NGA developer space is encouraged.

3.12.3 (U) Cloud Guidance

(U) The Intelligence Community has undertaken the IC Information Technology Enterprise (IC ITE) initiative as a platform for strengthening mission integration across the IC. IC ITE has designated service providers for cloud and common support services to "do in common what is commonly done". NGA has upheld its "all in" position by adopting community-provided services. Our objective is to provide GEOINT services for the community in a cloud-based enterprise.

(U) The IC has agreed that now is time to establish greater interoperability and efficiency within the community, and to implement an enterprise that will forge enduring synergies between agencies along with greater mission support and effectiveness. IC ITE has reached the stage where its enterprise services are fulfilling that sustainability goal of enabling IC interoperability and efficiency with the benefit of greater IC mission support.

(U) NGA Cloud Guidance provides Program Managers (PMs), developers, engineers, and organizational managers with a basic understanding of actions, constraints, and expectations of operations in the cloud environment. NGA Cloud Guidance provides prescriptive guidance and NGA-tailored industry best practices for cloud-based activities. The Annexes and White Papers are linked to references that provide more detailed technical implementation details important to architects, engineers, and developers.

(U) For purposes of guidance, an "activity" refers to any project, program, system, entity, instance, IT asset, or data that exists at NGA. It encompasses all software; hardware; purchased and NGA-produced data; and mission and business applications associated with the activity, whether funded by NGA or managed by NGA for others

(e.g., when we receive funds from others) to provide services on their behalf. Collectively, these terms are referred to as an “activity”.

(U) The mission objectives discussed in the GEOINT CONOPS 2022 will be achieved as we operate in the cloud. NGA will endeavor to operate all its enterprise activities in the cloud. NGA’s move to the cloud is one of many transformational initiatives that will occur on that path with the cloud being the foundation to enable that global mission support vision.

4 (U) ENGINEERING ROLES, RESPONSIBILITIES AND ENGINEERING GOVERNANCE

(U) This section describes how NGA systems engineering roles and responsibilities are executed across NGA.

4.1 (U) Roles and Responsibilities

4.1.1 (U) NGA Chief Engineer

(U) The NGA Chief Engineer (CE) is responsible for the execution and oversight of engineering activities, enterprise level decisions, and approvals for NGA. In this role, the CE approves and authorizes the engineering processes across the engineering lifecycle as captured in this SEP.

4.1.2 (U) Enterprise Engineering Office

(U) The Enterprise Engineering Office performs and provides enterprise-level oversight of the systems engineering functions and processes associated with the elements of engineering discussed in this SEP document. In addition to the elements discussed in this SEP, the Enterprise Engineering Office performs and/or supports activities such as Modeling and Simulation, Data Architecture, Analysis of Alternatives (AoAs)/Trade Studies, Legacy Retirement Planning, and Pre-Acquisition Support. The Enterprise Engineering Office leads the enterprise by defining the structures and patterns that will be followed and implemented.

(U) Enterprise Systems Engineering deals with services (which are components of a capability) as the smallest developmental unit (below which, things are acquired or created by programs or projects). This also means the NGA Enterprise Systems Engineering Office has the final responsibility for integration and end-to-end closure of NGA developed or acquired components into operations.

4.1.3 (U) Programs

(U) Programs are a directed activity to provide a new, improved, or continuing materiel, weapon, or information system in response to an approved need in accordance with an approved acquisition (cost, schedule, and technical baseline. For the purposes of this SEP, programs are defined and inclusive of MSAs, IPOs, Sensor Integration Program (SIP) Offices, Government Program Management Offices (GPMO), Programs, Integration Projects, Segments, and Teams.

(U) Each program is required to evaluate this SEP, determine gaps, and create supplemental documents (e.g., Program-Level SEP (PSEP) to ensure Systems Engineering needs are met (See PSEP template in Appendix C). PSEPs should include a discussion of the strategy for lifecycle readiness, transition to operations, mission assurance, and GEOINT protection. Programs also have a responsibility to develop

roadmaps specific to their mission needs with input from mission and technical stakeholders.

4.1.4 (U) Program Offices

(U) The Program Offices play a key role in enabling NGA's systems engineering activities and practices by steering support and development contractors towards compliance with the engineering practices and processes established by the NGA CE and the NGA Enterprise Engineering Office and other oversight entities. Program offices are also central to conducting user outreach that serves to enhance the engineering process by ensuring the user understanding of system capabilities and to ensure that mission needs are delivered. The user outreach engages users to identify critical intelligence problems to be addressed by systems capabilities and services.

4.1.5 (U) Development and Support Contractors

(U) The NGA SEP provides high-level guidance for contractor systems engineering activities. Besides the large-scale contracts for systems engineering and systems integration support, there are a variety of contracting vehicles/methods available to obtain contractor support on NSG systems engineering activities.

(U) The traditional aspect of the systems engineering relationship with development contractors is that government systems engineers and program managers review development contractor engineering implementation plans. However, there is a changing aspect of the systems engineering relation with contractors that is closely related to the move to agile acquisition. Up to the present time, the interactions between government systems engineers and contractor systems engineers have been based on relatively large, longer-term projects, and the focus was on traditional "requirements first" based approaches. Under agile acquisition, the focus is now on the outcomes of smaller, shorter-term tasks that target solutions based on delivering NSG capability in accordance with scheduled time-certain releases.

4.1.6 (U) Stakeholders

(U) The scope of the NGA SEP is intended to reflect the total technical approach taken by internal and external stakeholders to include government and contractors supporting the NSG, and to support the interaction of the NSG with external organizations and communities. The systems engineering functions include key roles supporting the integration of the NSG with larger communities outside of NGA such as the IC and DoD. Additional roles where NSG systems engineers support external organizations includes definition of needed services, discovering available services, and representing NGA interests when determining community plans for new technology adoption such as cloud computing and data center consolidations.

4.2 (U) Engineering Governance

(U) The NGA Chief Engineer uses the NGA ERB as the mechanism to execute enterprise-level oversight over NGA's engineering processes and execution, and the processes as defined in this SEP.

(U) The NGA ERB is an enterprise-focused forum to ensure engineering activities are integrated, orchestrated, harmonized, and prioritized across NGA. NGA ERB results and outcomes facilitate and drive the following:

- Decisions pertaining to actionable engineering and architecture plans
- Consistency across and among engineering efforts
- Alignment to NGA enterprise priorities and objectives
- Coordination, Integration, Collaboration and Visibility

(U) The NGA ERB is a decision forum to ensure engineering, analyses, solutions, and recommendations are aligned, consistent, and support enterprise priorities and objectives. The NGA ERB reviews and approves engineering deliverables, engineering implementation approaches and solutions, and recommendations that impact the NGA enterprise, the NSG and/or the Allied System of Geospatial Intelligence (ASG), and NGA Mission Partners.

5 (U) ENGINEERING TOOLS

(U) NGA has implemented a host of tools to support the systems engineering processes. Descriptions of those tools is provided below.

(U) Baseline Management System (BMS) - The BMS is an integrated application suite built on ServiceNow and hosted on C2S. The BMS functionality includes applications to manage the agency's RFC workflow, the CM Library for all Configuration Control Board controlled Configuration Items (CIs), the NGA Effectivity Log, to include Effectivity requests and modifications. The tool enables NGA personnel and contractors to perform RFC and Engineering Change Proposal (ECP) activities online to document changes to the IT baseline, coordination of these changes and recording of comments, recommendations, and costs inputs associated with the proposed changed. The BMS is the repository for forms, templates, as well as CM related training material and RFC library. The tool provides an abstract of the purpose of each document, points of contact for the document, a list of Change Notices, status on recent updates, last date published, and a link to the complete document in BMS. This information provides the RFC records with current versions of documents for impacts and to see the status of each document relative to currency of updates.

(U) DOORS Next Generation (DNG) – The DNG is a requirements analysis tool used to establish and maintain lifecycle requirements traceability from funding source to execution. Analysis of DNG data and established links enable decision-makers to identify gaps in coverage, project expenditures, and eliminate waste.

(U) Enterprise Schedule Management System (ESMS) - The ESMS is comprised of a SQL Database, SQL Server Reporting Service (SSRS), Microsoft Project Server, and SharePoint. The contents and schedules hosted in the ESMS can be accessed via MS Project Professional, Project Web App (PWA), or the Schedule Visualization and Reporting Tool (SVRT).

(U) Schedule Visualization and Reporting Tool (SVRT) - The SVRT provides visibility to the NGA Configuration Control Board Managed Schedule (NMS) and other schedules hosted in the ESMS.

(U) Fit-for Purpose Standards Selection Tool (FfP) - Hosted in the Advanced Technical Leadership Analytic Suite (ATLAS), the FfP provides a rational means to identify an initial set of standards based on the JARM's Taxonomy Services Level, and using its Taxonomy Service Categories (TSCs) to select standards. Each TSC is mapped to a set of standards, either IT and/or GEOINT. The tools within the FfP enable a down-selection of standards present in the Enterprise Standards Baseline (ESB) to a critical list of standards, which satisfy most of the needs of the program or project.

(U) Jira™ – Atlassian's Jira is a project management and issue-tracking tool for modern software development. Jira provides the necessary functions for planning, prioritization, tracking, reporting, and collaboration. Jira also provides a Kanban Board capability that gives project teams the ability to track what tasks need to be completed next and by when. Basic dashboards can be built directly in Jira or, using the Jira to Tableau connector app, Jira can feed data to Tableau to help create complex data visualization dashboards.

(U) Confluence™ – Atlassian's Confluence capability provides a team workspace where knowledge can be shared. Documents, templates, processes, and procedures can be posted in the Confluence workspace. It is also as collaboration site where documents can be created and are editable. Changes to the document are visible and multiple team members can be working on the document at the same time. Comments can be captured and there is a log of who and when the comments was made. Confluence provide a team calendar space, blogging capabilities, a place to create and maintain meeting notes and a means to capture Jira reports.

(U) Tableau™ - Tableau provides visualizations of complex data. Tableau can be used to build automated dashboards for visualizing progress, trends, and status. Tableau reports are drag and drop rather than hardcoded. With the "Tableau Connector for Jira" Program and Solution Epic data can be pulled from Jira and provides the capability to visualize epic/requirement hierarchy, and the status of PEs and SEs development efforts at any program level. Reports can be created to provide meaningful visualizations and incorporate into program dashboards for the enterprise. Within these dashboards, various organizations utilize the pulldown capability to focus on their area of interests. These visualizations are used to convey integration, status and progress information at all levels of the organization in the same format across programs.

(U) Readiness Event Tracker (RET) - NGA uses the RET as the authoritative data repository for capturing, storing and displaying information and status of upcoming and planned; as well as accomplished readiness events.

(U) Cameo Systems Modeler - Cameo Systems Modeler is a robust Model-Based Systems Engineering (MBSE) tool, which strictly enforces most of OMG SysML's syntax and semantics, and offers support for basic requirements traceability, intermediate model-based simulation and automated document generation. NGA currently portrays architectural views and diagrams using Cameo Systems Modeler in accordance with (IAW) the Unified Profile for DoDAF and MoDAF (UPDM), but will be migrating to the Unified Architecture Framework (UAF) upon upgrading to Cameo Enterprise Architecture. Design patterns, and stylistic guidance can be found in the TAED Cameo Style Guide

6 APPENDIX A – (U) ACRONYMS

UNCLASSIFIED	
Acronym	Definition
ANSI	American National Standards Institute
API	Application Programming Interface
ART	Agile Release Train
ASG	Allied System of Geospatial Intelligence
ASP	Application Service Provider
ATLAS	Advanced Technical Leadership Analytic Suite
AO	Acquisition Office
AoA	Analysis of Alternatives
ATO	Authority to Operate
ASoT	Authoritative Source of Truth
BMS	Baseline Management System
CAE	Component Acquisition Executive
CBA	Capabilities Based Assessment
CCA	Configuration Control Authority
CCB	Configuration Control Board
CDD	Capability Development Document
CE	Chief Engineer
CI	Configuration Item
CIO	Chief Information Officer
CIO-T	CIO and IT Services Directorate
CM	Configuration Management
CNSS	Committee on National Security Systems
CONOPS	Concept of Operations
COR	Capability Oriented Requirements
COTS	Commercial Off-The-Shelf
CRR	Consolidated Reference Repository
CV	Capability View
DAG	Defense Acquisition Guidebook

UNCLASSIFIED	
Acronym	Definition
DAU	Defense Acquisition University
DD	Design Document
DE	Digital Engineering
DepSO	Deputy Standards Officer
DGIWG	Defense Geospatial Information Working Group
DISR	DoD IT Standards Repository
DNG	DOORs Next Generation
DNI	Director National Intelligence
DoD	Department of Defense
DoDAF	DoD Architecture Framework
DOORS	Dynamic Object Oriented Requirements System
DSP	Defense Standardization Program
EA	Enterprise Architecture
ECP	Engineering Change Proposal
ER	Enterprise Readiness
ERB	Engineering Review Board
ERM	Enterprise Release Management
ERP	Enterprise Readiness Process
ESB	Enterprise Standards Baseline
ESM	Enterprise Schedule Management
ESMS	Enterprise Schedule Management System
FEA	Federal Enterprise Architecture
FfP	Fit for Purpose
FRPE	Final Release Planning Event
GEA	GEOINT Enterprise Architecture
GEOINT	Geospatial Intelligence
GFMSA	GEOINT Functional Manager Standards Assessment
GI	Geospatial Intelligence
GOTS	Government Off-The-Shelf
GPMO	Government Program Office
GWG	GEOINT Working Group
IA	Information Assurance

UNCLASSIFIED	
Acronym	Definition
IC	Intelligence Community
ICBD	Independent Cost Baseline Document
ICCR	Intelligence Community Capabilities Requirements
ICD	Intelligence Community Directive
IC ITE	IC IT Enterprise
IC PAG	IC Program Architecture Guide
IEEE	Institute of Electrical and Electronics Engineers
IETF	Internet Engineering Task Force
IPO	Integrated Program Office
ISO	International Organization for Standardization
ISP	Infrastructure Service Provider
IT	Information Technology
ITRM	IT Risk Management
ITPOM	IT Program Objective Memorandum
JARM	Joint Architecture Reference Model
JCIDS	Joint Capabilities Integration and Development System
JIE	Joint Information Environment
LBC	Lean Business Case
MBSE	Model-Based System Engineering
MDA	Milestone Decision Authority
MDA	Model Driven Architecture
MSA	Major Systems Acquisition
MVP	Minimum Viable Product
MVPf	Minimum Viable Product function
NAS	NSG Applications Schema
NATO	North Atlantic Treaty Organization
NCCB	NGA Configuration Control Board
NCMP	NGA Configuration Management Plan
NCOW	Net-Centric Operations Warfare
NEC	NSG Entity Catalog
NERF	NGA Enterprise Request Framework
NERP	NGA Enterprise Requirements Process

UNCLASSIFIED	
Acronym	Definition
NGA	National Geospatial-Intelligence Agency
NIS	NGA Integrated Schedule
NIST	National Institute of Standards and Technology
NMS	NGA Configuration Control Board Managed Schedule
NRAI	NSG Requirements for Application and Infrastructure
NSG	National System for Geospatial-Intelligence
NSMP	NGA Schedule Management Plan
NSS	National Security System
NTO	NGA Test Organization
OAR	Operational Acceptance Review
ODNI	Office of the Director of National Intelligence
OGC	Open Geospatial Consortium
OPR	Office of Primary Responsibility
ORR	Operational Readiness Review
OSO	Office of Strategic Operations
OV	Operational View
PE	Program Epic
PIS	Public Interface Specification
PM	Program Manager
PMO	Program Management Office
POC	Point of Contact
POM	Program Objectives Memorandum
POPM	Product Owner/Product Manager
PSEP	Program-Level Systems Engineering Plan
PSM	Platform Specific Models
PWA	Project Web App
RD	Requirements Document
RET	Readiness Event Tracker
RFC	Request for Change
RIO	Risks, Issues, and Opportunities
RPE	Release Planning Event
RPM	Required Performance Measure

UNCLASSIFIED	
Acronym	Definition
SAFe®	Scaled Agile Framework®
SAM	Software Asset Management
SDLC	System Development Life Cycle
SE	Solution Epic
SEMP	System Engineering Management Plan
SEP	Systems Engineering Plan
SIA	Special Interest Acquisition
SIDR	Security Impact Decision Request
SIP	Sensor Integration Program
SRD	System Requirements Document
SRR	System Requirements Review
SOC	Statement of Capability
SOP	Standard Operating Procedures
SoS	System of Systems
SPID	Security Plan Identification
SS	Service Specification
SSRS	SQL Server Reporting Service
STANAG	NATO Standards Agreement
StdV	Technical Standards View
SV	System View
SvcV	Service View
SVRT	Schedule Visualization and Reporting Tool
SWAP	Software Approval Process
TAC	Integration Test & Image Quality Office
TBD	To Be Determined
TBR	To Be Resolved
TC	Technical Committee
T&E	Test and Evaluation
TEM	Technical Exchange Meeting
TEMP	Test and Evaluation Master Plan
TRR	Test Readiness Review
TSC	Taxonomy Service Categories

UNCLASSIFIED	
Acronym	Definition
TST	Technical Service Taxonomy
UAF	Unified Architecture Framework
UFS	User Facing Services
UPDM	Unified Profile for DoDAF and MoDAF
URL	Uniform Resource Locator
WAND	Web-Assisted NGA Documentation
WG	Working Group
W3C	World Wide Web Consortium
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**7 APPENDIX B – (U) MSA MILESTONE DOCUMENTATION REQUIREMENTS, ICS
801-01**

UNCLASSIFIED		
Initial Concept Studies Phase	Phase A	Phase B
Analysis of Alternatives (AoA)	Acquisition Strategy	Acquisition Strategy
ICBD, POE, ICE & Affordability Assessment	Program Management Plan (PMP)	Program Management Plan (PMP)
Technology Development Strategy (TDS)	Bandwidth Capability Assessment	Systems Engineering Plan (SEP)
CONOP	Vulnerability Assessment Report	Test and Evaluation Master Plan (TEMP)
Program Architecture	Systems Engineering Plan (SEP)	Risk Identification & Mitigation Plan
Initial Program Management Plan	Test and Evaluation Master Plan (TEMP)	ICBD, POE, ICE & Affordability Assessment
Security Classification Guide	CONOP	Integrated Program Assessment
	Risk Identification & Mitigation Plan	IC Enterprise Architecture Compliance
	ICBD, POE, ICE & Affordability Assessment	Information Support Plan
	Integrated Program Assessment	Security & Counter Intelligence Assessment
	IC Enterprise Architecture Compliance	Verification that key assumptions remain valid
	Information Support Plan	Security Classification Guide
	TDS & Technology Maturity Assessment	
	Security & Counter Intelligence Assessment	
	Verification that key assumptions remain valid	
UNCLASSIFIED		

(U) Analysis of Alternatives (AoA) - is an analytical comparison of the operation effectiveness, suitability, and Life-Cycle cost of alternatives that satisfy established capability needs. Initially, the Analysis of Alternatives process typically explores numerous conceptual solutions with the goal of identifying the most promising options, thereby guiding the Concept Refinement Phase. Subsequently, at Milestone B (which usually represents the first major funding commitment to the acquisition program), the Alternative of analysis is used to justify the rationale for formal initiation of the acquisition program.

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(U) AoA Output: Options for prototypes

(U) ICBD / ICE / Life Cycle Cost Estimate – Estimated Monetary Cost

(U) Concept of Operations (CONOPS) – The CONOPS is a description of how a system or collection of processes are anticipated to be performed. DoDAF operational views and system views complement the CONOPS and describe the system in greater detail.

(U) CONOPS Output: Inputs to the Solution Epic LBC(s)

(U) Program Architecture - Are the required IC PAG artifacts

(U) Program Management Plan (PMP) - For Major System Acquisitions (MSAs) entering Phase A, a "Phase A" PMP is established that includes cost, schedule, and performance objectives for Phase A. Reviews of the Phase A PMP will focus on progress toward completing the documentation (e.g., Phase A exit criteria, validated and approved requirements document for Phase A, Technology Maturity Objectives), and other elements necessary to enter Phase B. The PMP baseline for an MSA is established at Milestone B. For MSAs entering Phase B, the PMP addresses the total acquisition and is focused on the cost, schedule, and performance goals associated with the development, integration, and demonstration of the acquisition. For MSAs entering Phase C, the PMP also addresses the total acquisition and is focused on the cost, schedule, and performance goals associated with the production and fielding aspects of the acquisition.

(U) System Engineering Plan (SEP) – describes technical approach, including processes, resources, and metrics. Details timing and conduct of SE technical reviews.

- **(U) SEP Output:**
 - Scope of System Development
 - Schedule Management – effectivities/readiness & technical reviews
 - Standards impacting program - NFRs
 - Stakeholders

(U) The Acquisition Strategy – The Acquisition Strategy is a comprehensive plan that identifies and describes the acquisition approach that Program Management will follow to manage program risks and meet program objectives. The Acquisition Strategy guides program execution across the entire program life cycle and is updated at every major milestone and review. An approved plan is required by DoD Instruction 5000.02 “Operation of the Defense Acquisition System” prior to Milestone A and before any Request for Proposal (RFP) release for the Engineering and Manufacturing Development (EMD) phase. The acquisition strategy is also updated at Milestone B, Milestone C and Full Rate Production Decision (FRPD) review.

(U) Technology Development Strategy (TDS) - A preliminary description of how the potential acquisition will be divided into increments based on mature technologies; a preliminary acquisition strategy to include overall cost, schedule, and performance goals; specific cost, schedule, and performance goals, including exit criteria, for Phase A; the approach for management of data assets, a list of known or probable Critical Program Information and potential countermeasures; and a time-phased workload assessment.

(U) Test and Evaluation Master Plan (TEMP) – The TEMP describes program unique IV&V activities in addition to processes and procedure described in the NGA Test Evaluation Master Plan (TEMP), these can be the use of DevSecOps, the SCP, the testing readiness review process, etc.

(U) Risk Identification & Mitigation Plan - The Risk Management Plan (RMP) describe the program’s unique ROM activities and the interaction with NGA’s ROM process in addition to NGA’s established CM processes and procedures

(U) Independent Program Assessment – An IPA is a focused, short-duration, peer review performed by experienced technical experts and acquisition professionals. The objective of an IPA is to assess the adequacy of the proposed approach and an acquisition's readiness to proceed including a technology maturity assessment. The IPA focus is the identification and evaluation of key acquisition elements, with the exception of cost, since the Independent Cost Estimate (ICE) addresses cost-related elements of an acquisition in parallel with IPA.

8 APPENDIX C – (U) APPLICABLE DOCUMENT AND TOOLS LINKS

(U) Addresses and links to applicable directives, instructions, and documents that are referenced in this SEP are provided below.

GEA	(U) http://gea-baseline.nga.ic.gov/gea_baseline.html
NCMP, NERP, NGA TEMP, NSMP, ITRMP	(U) https://bmwiz.nga.ic.gov/ (BMS Library)
DAG	(U) dau.edu/tools.dag
DoDD 5000.01	(U) https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodd/50000p.pdf
DoDI 5000.2	(U) https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500002p.pdf
DoDI 5000.87	(U) https://intelshare.intellink.gov/sites/micp/Landing/IUS_Implementation_DocumentLibrary/IUS%20Accounting%20Guidance/DODI%20500087.pdf#search=DODI%205000%2E87
DoD DE Strategy Guide	(U) ac.cto.mil/wp-content/uploads/2019/06/2018-Digital-Engineering-Strategy_Approved_PrintVersion.pdf
ICCR (ICD115)	(U) https://www.intranet.nga.ic.gov/org/d/dd/Acquisition%20Policy%20Documents/Policies,%20Procedures%20and%20Guidances/IC%20Policies/ICD%20115_21Dec2012.pdf#search=ICD%20115
JARM	(U) https://intellipedia.intellink.ic.gov/wiki/Joint_Architecture_Reference_Model#.28U.29_IC_JARM_v5.0_Release
ICD 801	(U) ICD 801.pdf
IC PAG	(U) IC PAG.pdf
ODNI ICS 801-01	(U) ICS 801-01.pdf (U)
Confluence	(U) https://usms.nga.ic.gov/confluence/
Jira	(U) https://usms.nga.ic.gov/jira/
VOLT	(U) https://volt.nga.ic.gov/rctt/rctt_extjs_external.cfm

**9 APPENDIX D – (U) PROGRAM-LEVEL SYSTEMS ENGINEERING PLAN (PSEP)
TEMPLATE**

(U) INSERT PROGRAM NAME

**(U) PROGRAM-LEVEL SYSTEMS
ENGINEERING PLAN
(PSEP)**

Version xxx

Day Month Year

Abstract: This document describes the conduct, management, and control of the systems engineering effort for the acquisitions for (insert name of program)

Key Words: Acquisition, Activity, Allocation, Architecture, Interface, Management, (name of program), Performance, Quality, Requirements, Integration, Configuration Management, Review, Risk, Service, System, Systems Engineering, Information Assurance, Test, Standards, Transition, Validation, Verification

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List of Figures, If Needed

Sample Revision/Change History Table, if Required

REV	Date	Items Affected	Remarks/Change Authority
Rev B	DD MM YY		

XXX Program Overview

(U) Introduction and Purpose

(U) Sample Language: Describe the primary purpose of the XXX Program-Level Systems Engineering Plan (PSEP) is to describe the systems engineering processes in the XXX Program supporting delivery of capabilities to both the Intelligence Community (IC) and the Department of Defense (DoD).

(U) The XXX program systems engineering processes, a hybrid of agile and traditional. Note that the important engineering functions are retained; however, the flexible and non-sequential nature of the XXX program agile acquisition management is emphasized (insert applicable figures, if desired)

1. (U) Applicable Documents

(U) Sample Language: The XXX Program PSEP is consistent with IC, DoD and NGA Acquisition and Systems Engineering Management Directives, Instructions and Guides, and the NGA SEP (programs evaluate governing directives and make reference to them here). When appropriate, the XXX Program system engineering processes may be tailored, as needed. Refer to Appendix XXX, Applicable Documents, for the complete list of applicable documents.

2. (U) Program Systems Engineering Management Processes

Describe who, what, when and why. For example, if it is an MSA, say so, if not tell what the program is and why it is being executed. At a high-level say what problems/capability gaps is this program trying to address describe the high-level systems engineering methodologies is using and how they are managed.

3. (U) Technical Activities Summary

(U) Description: This section summarizes the XXX systems engineering technical activities necessary to deliver XXX capabilities. The section is applicable to systems engineering activities being performed at the Strategic, Enterprise, and Program levels. This is where the NGA program captures program-specific systems engineering activities, if not already addressed in NGA's Management Plan documentation (NGA's Systems Engineering, Requirements Management, Configuration Management, Enterprise Schedule Management, Risk Management, Test and Evaluation Master Plans).

3.1 (U) Systems Engineering Activities

- (U) Describe the Agile, or Traditional, or Hybrid systems engineering processes being used and are managed. Example are:
 - Describing the management of XXX Program Increment Planning events and execution
 - Describing how cross-program agile development activities are being coordinated

- (U) Defining what Data Standards and Mandated Standards are to be employed and how the baseline is going to be managed
- (U) Defining how the Capabilities Roadmap is developed and maintained. The Capabilities Roadmap defines the necessary capabilities needed to be developed to meet program milestones, Enterprise Effectivities, or both.
- (U) Defining program. Cross-program Integration Management processes
- (U) Defining cloud strategy planning and implementation, if required
- (U) Defining use of Service-Centric Platform (SCP) activities, if required
- (U) Defining Enterprise-level program integration, test, verification and validation activities:
 - Leading XXX program Test and Verification events; support Testing Readiness Review activities
 - Verifying satisfaction of xxx program requirements and validate delivery of MSA capabilities (if an MSA)

3.2 (U) Program Requirements Management Process

(U) **Describe** how XXX program Requirements Management process translate creates a full set of allocated and verifiable XXX program requirements for the acquisition office and developers.

(U) **Suggested items for inclusion.** Explain how XXX program:

- Develops and maintains XXX program requirements, to include attributes defining traceability and verification methods
- Derives and decomposes XXX program requirements to XXX and allocates to program acquisition contracts
- Capability gap identification
- Evaluates recommended changes against the requirements baseline
- Assesses requirements with respect to security and information assurance controls
- Assesses and allocates Infrastructure requirements
- Documents and maintains the requirements baseline in appropriate tools/repositories

3.3 (U) Program Architecture Management

(U) **Describe** XXX program Architecture development process. What are architecture artifacts are being developed and how. Also what database is used to house the artifacts, if applicable?

3.4 (U) Integration, Verification, and Validation Activities

(U) **Describe** XXX program unique IV&V activities in addition to processes and procedure described in the NGA Test Evaluation Master Plan (TEMP).

(U) Describe XXX program interactions with the NGA Test Organization (NTO), use of the DevSecOps capabilities and usage of the Service-Centric Platform (SCP), if applicable.

3.5 (U) Configuration Management (CM)

(U) Describe program program-unique CM activities in addition to NGA's established CM processes and procedures.

3.6 (U) Risk and Opportunity Management (ROM)

(U) Describe XXX program unique ROM activities and the interaction with NGA's ROM process in addition to NGA's established CM processes and procedures.

(U) Sample Language: XXX program Risk and Opportunity activities include:

- Development of a plan for the management of Risks and Opportunities
- The identification of, documenting, and validating Risks and Opportunities to determine what eventualities are likely to affect the XXX program efforts
- Assessing and analyzing validated Risks and Opportunities by determining the probability of occurrence and the performance, schedule, and/or cost impact using technical details, root cause analysis, engineering judgment, and lessons learned
- Executing actionable tasks within the Response/Handling Plan and track through to completion against planned and actual end dates
- Monitoring and reporting of risk status to ensure the progression of risk mitigation and opportunity exploitation activities to successful completion

3.7(U) Life Cycle Readiness

(U) Describe XXX program's unique Life Cycle Readiness processes. The NGA Tier 2 Readiness Process is tailorable and is applicable to both Traditional and Agile development programs. The type of documentation required for each milestone, or program review is also tailorable. If the program is an MSA describe what artifacts, e.g., PMP, PSEP, TEMP, etc., are to be provided at each Major Milestone and the level of maturity required.

(U) Program Life Cycle Readiness activities may include:

- Pre-Acquisition Readiness
- Initial Design Readiness
- Development Readiness
- Deployment Readiness
- Prototype Readiness
- Sustainment Readiness

3.8 (U) Schedule Management

(U) Describe any program-unique Schedule Management activities that are not prescribed in NGA's Enterprise Schedule Management plan. The XXX program schedule should include dates for the funded and planned xxx program capabilities deliveries,

dependencies on internal and external program, milestones, major testing and integration activities and Effectivities.

(U) 3.9 (U) Technology Insertion Process

Describe the XXX program actions by XXX program systems engineers to mature new capabilities, and manage the insertion and integration of the capabilities at all levels of the enterprise.

3.10 (U) Information Assurance

In addition to referring to the NGA SEP, section 3.7 (U) Security Engineering Objectives to guidance on Security Engineering approach, please describe any usage of the DevSecOps pipe line.

3.11 (U) Discrepancy Reporting and Resolution

(U) Define the XXX program's Discrepancy, Reporting, and Resolution processes

4. (U) Roles and Responsibilities

(U) Describe the XXX program *Systems Engineering Roles and Responsibilities*; capture key roles and responsibilities in implementing the XXX program engineering processes. As needed, reference NGA SEP for additional information on all NGA SE roles and responsibilities.

Example Table XXX: (U) XXX Program Systems Engineering Roles and Responsibilities

Role		Responsibilities
XXX Program Manager	•	- Approval authority for all XXX program configuration items, including SE products, unless delegated - Approves ROMB
XXX Deputy PM	•	Acts for XXX PM, as delegated
XXX Chief Engineer	•	•
VSE	•	•
Product Manager	•	
Product Owner	•	

Role		Responsibilities
ISSE	•	- Ensures compliance of all technical SE efforts with current security policies, guidance, and standards - Performs security review of SE products as required
Lead Integrator	•	•

5. (U) Key Products

Captures key XXX program SE Products produced and used during implementation of the XXX program engineering methodology. Program may leverage existing NGA Enterprise-level documentation, if appropriate. This list is not intended to be all inclusive.

Example: Table XXX: (U) Key XXX Program Systems Engineering Products

	Systems Engineering Products
Program SEP	
Architecture Artifacts	• Refer to NGA Architecture Style Guide to what is required
Requirements Management	• Refer to NERP
Technology Insertion	• Technology Transition Strategy
Schedule	• XXX Program Master Schedule
Life Cycle Readiness	• Readiness Assessment Briefings and Evidence
Integration, Verification, Validation	• Test & Verification Plans and Procedures
XXX Program Configuration Management	
Risk and Opportunity Management Plan	• Risk/Opportunity Jira Database
Information Assurance	- XACTA Database Entries - Plan of Actions & Milestones
Discrepancy Management	- DR assessments and technical investigations - DR resolutions and fixes

6. (U) Key Forums/IPTs/Working Groups

List xxx Program-unique forums, IPTs, and working groups

Appendix A: (U) Acronym List (Unique to XXX Program)

Acronym	Definition

Appendix B: (U) Applicable Documents

Document	Document Title